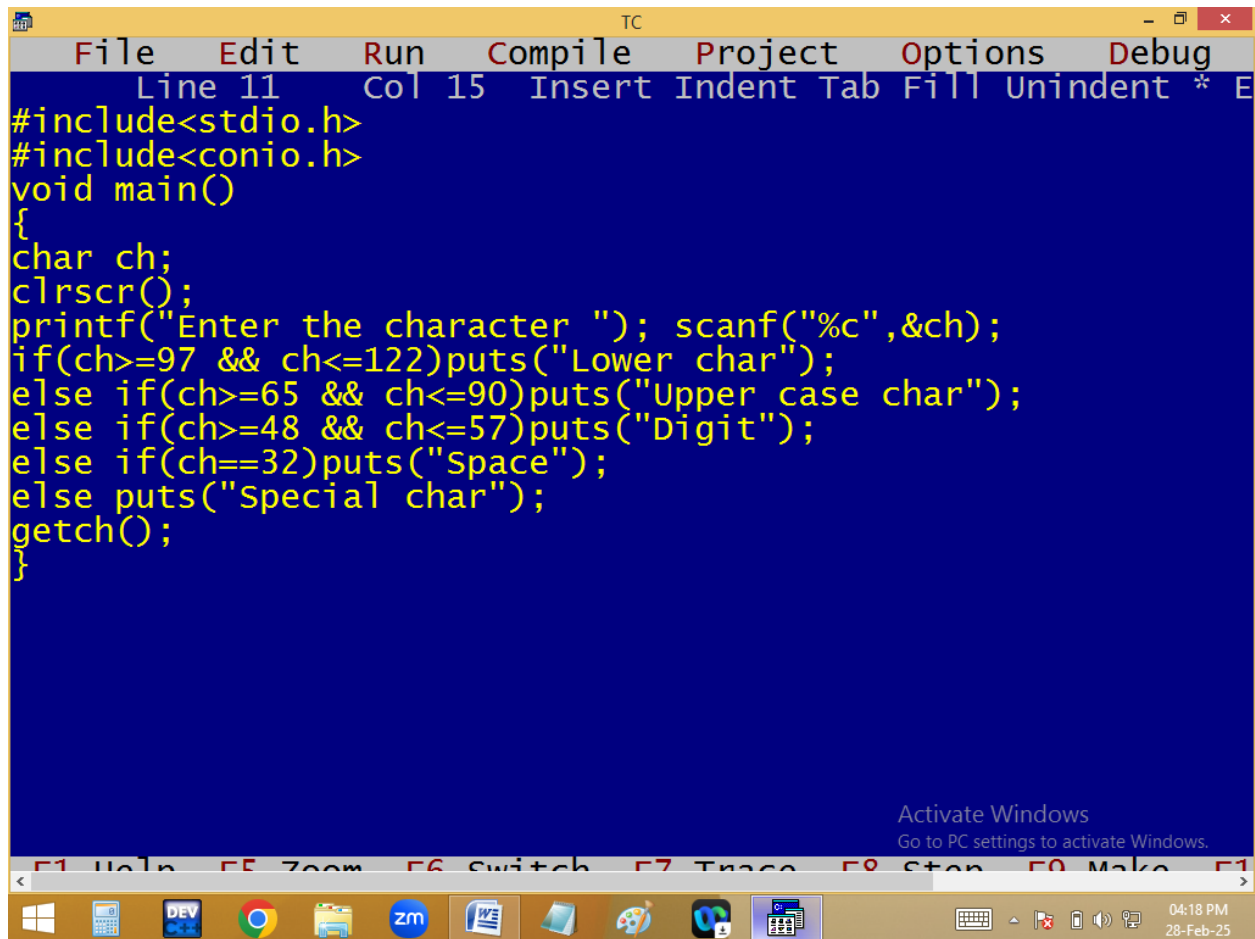


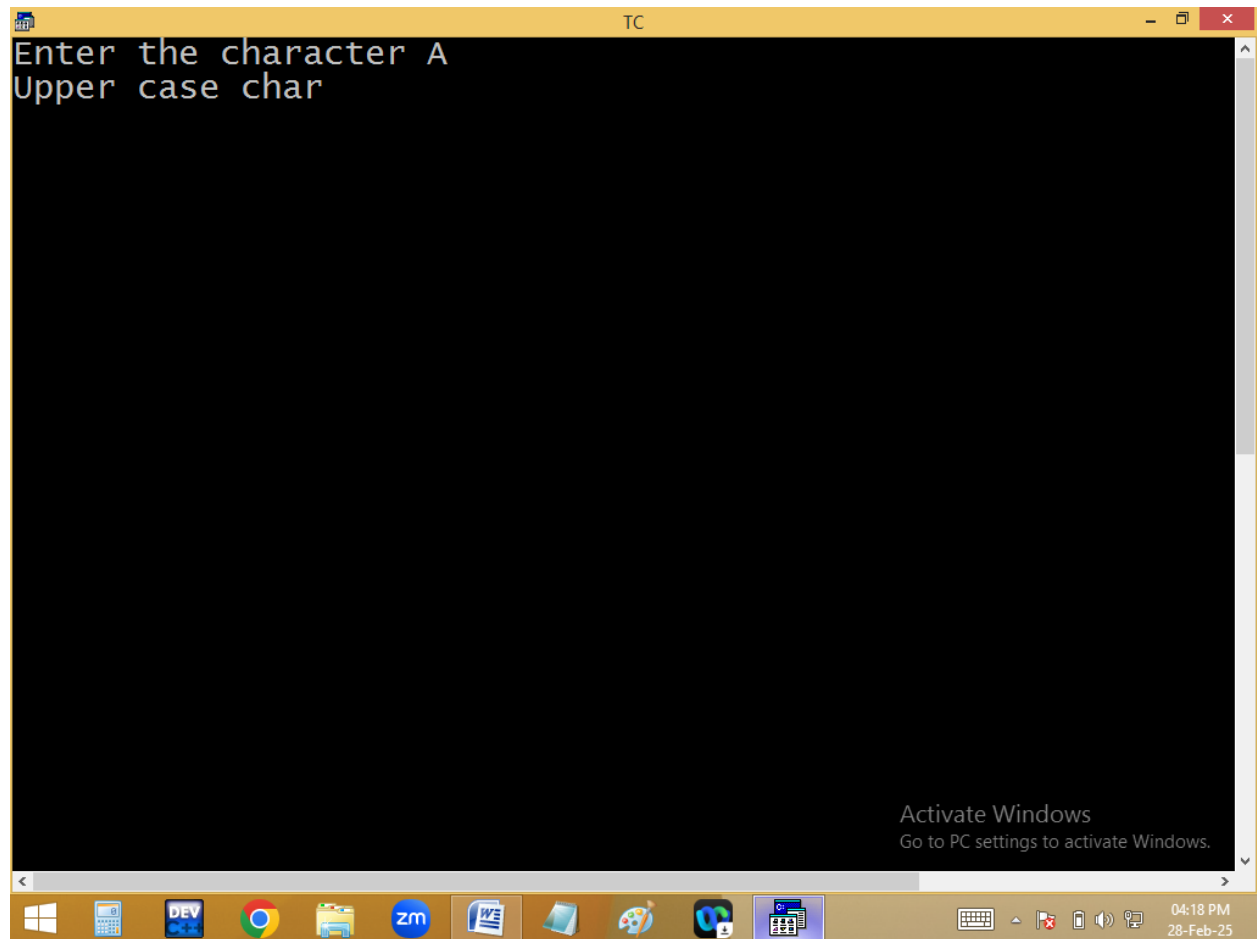
**Finding lower / upper / digit / space / special char with ascii values:**

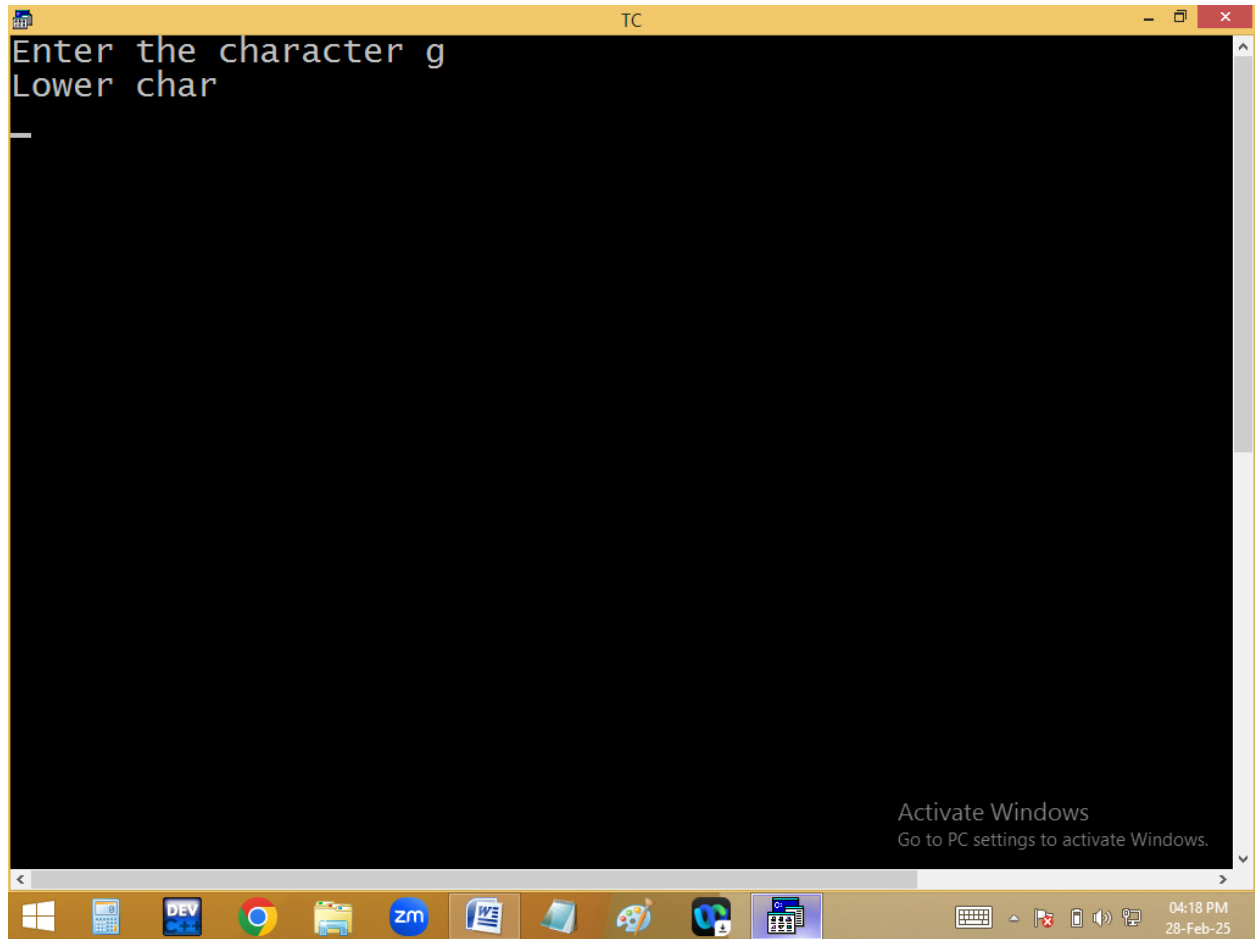


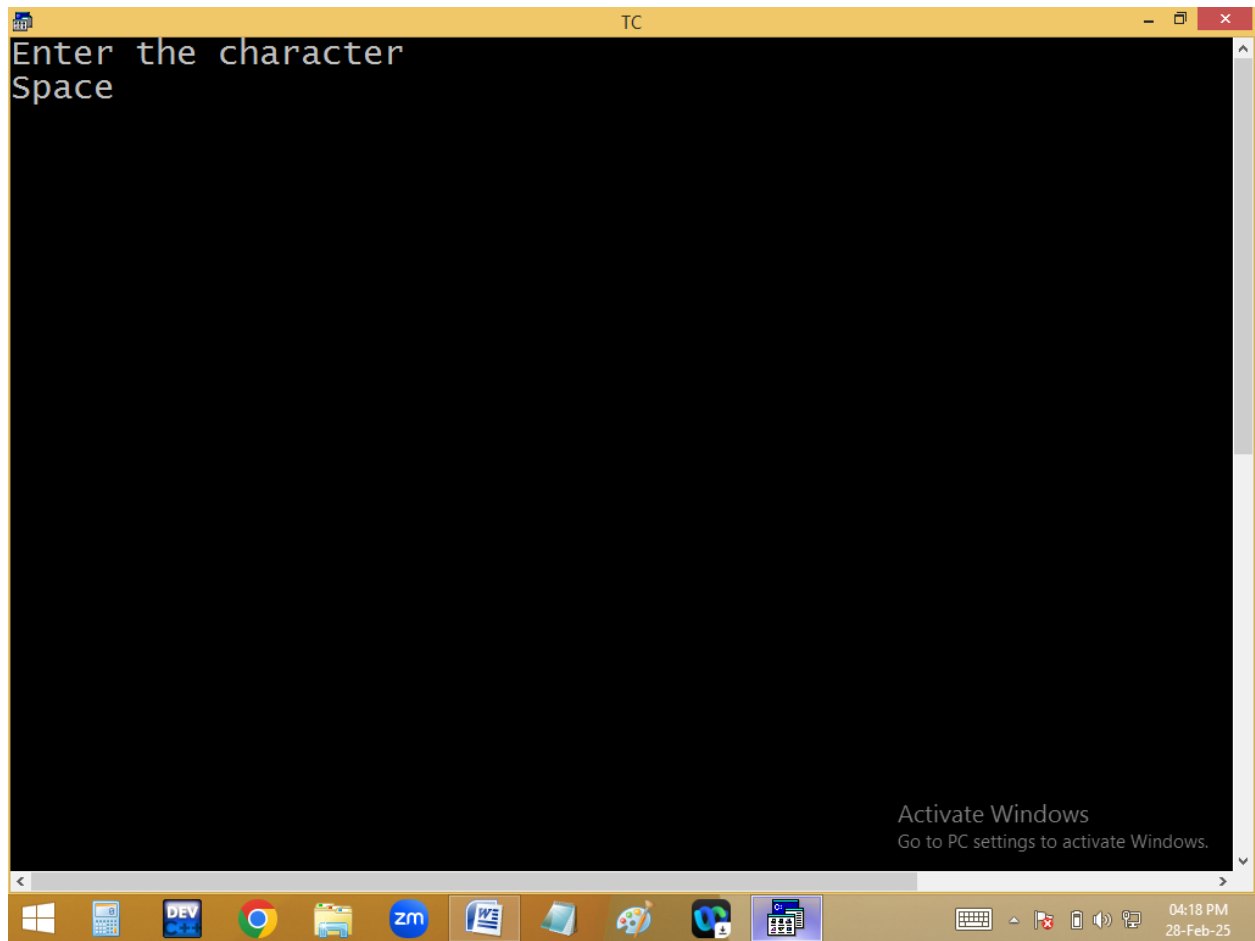
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 11", "Col 15", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "\*", and "E". The main editing area has a dark blue background with yellow text. The code is as follows:

```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>=97 && ch<=122)puts("Lower char");
else if(ch>=65 && ch<=90)puts("Upper case char");
else if(ch>=48 && ch<=57)puts("Digit");
else if(ch==32)puts("Space");
else puts("Special char");
getch();
}
```

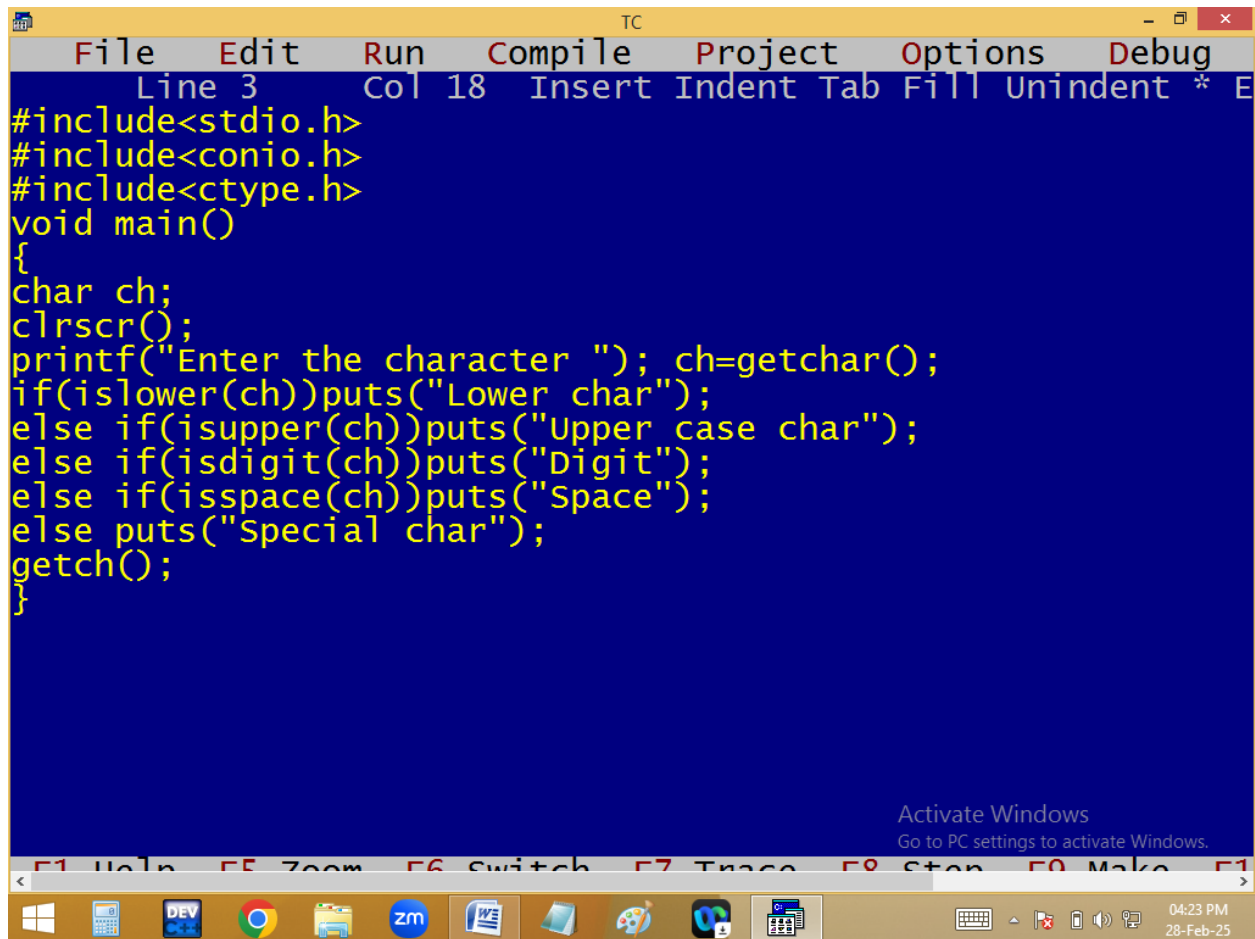
At the bottom of the window, there is a taskbar with various application icons including Windows Explorer, Google Chrome, and a calculator. On the right side of the taskbar, the system clock shows "04:18 PM" and the date "28-Feb-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.







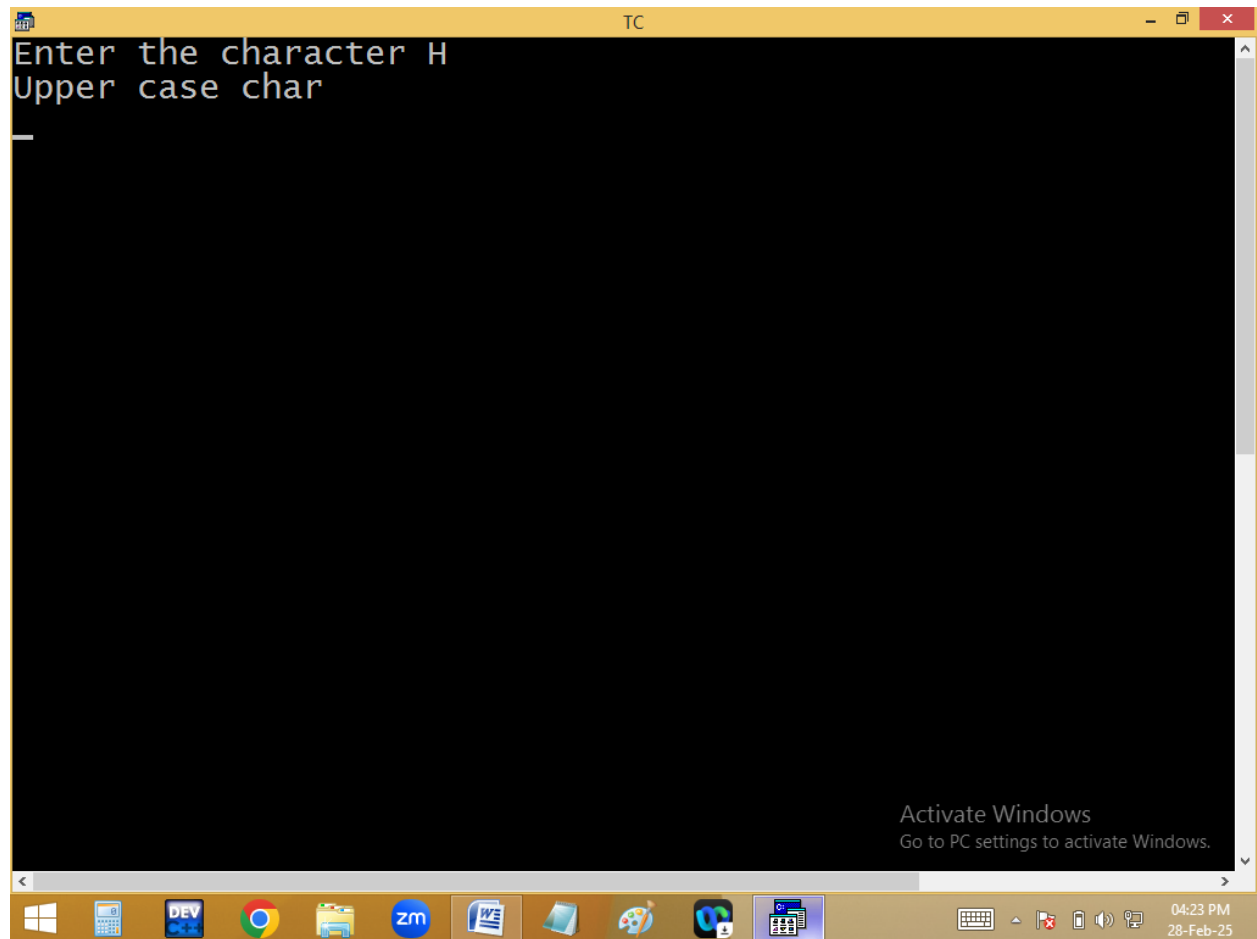
**Using predefined functions:**

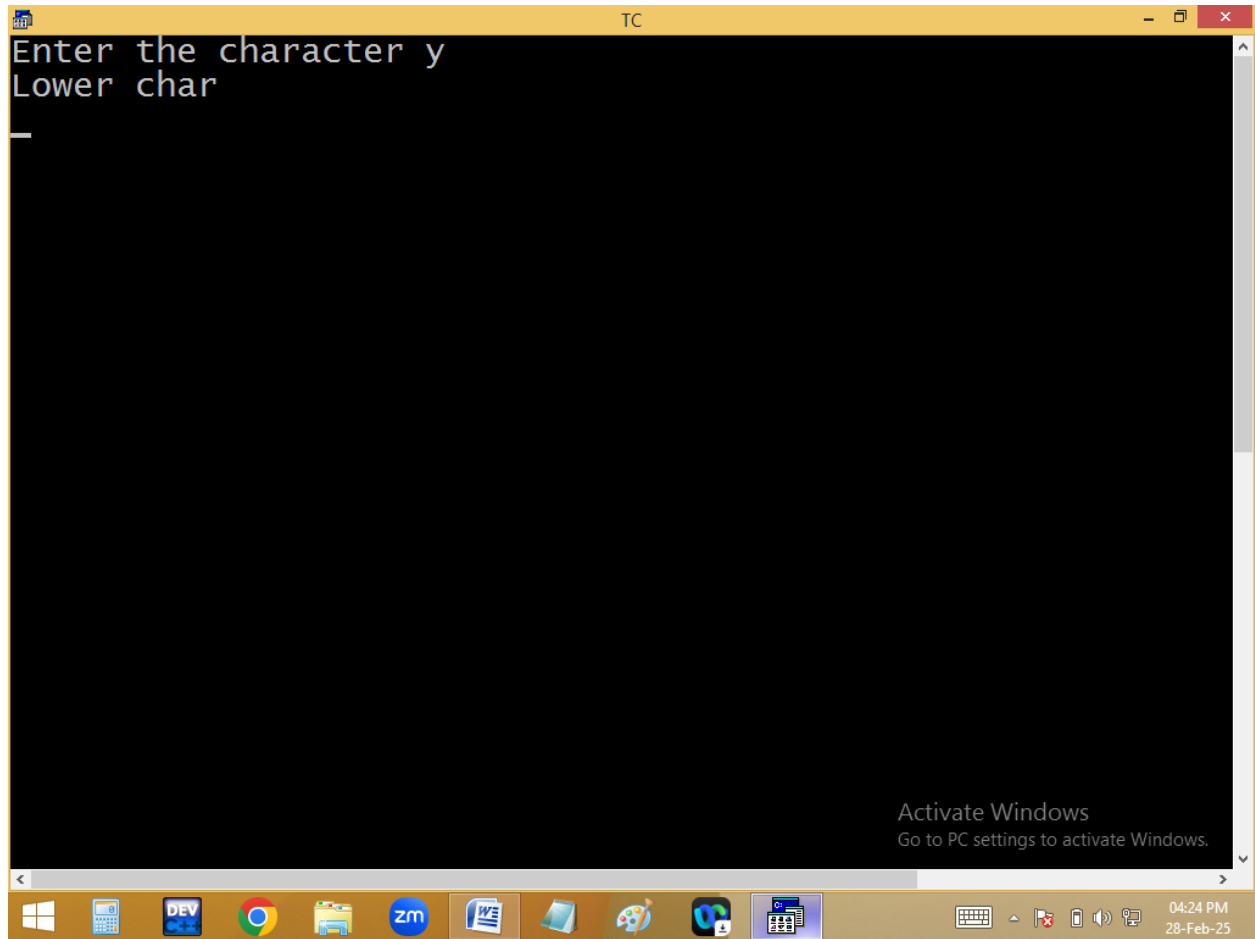


The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, the status bar shows "Line 3", "Col 18", and a list of keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "\*", and "E". The main editing area has a dark blue background with yellow text. The code is as follows:

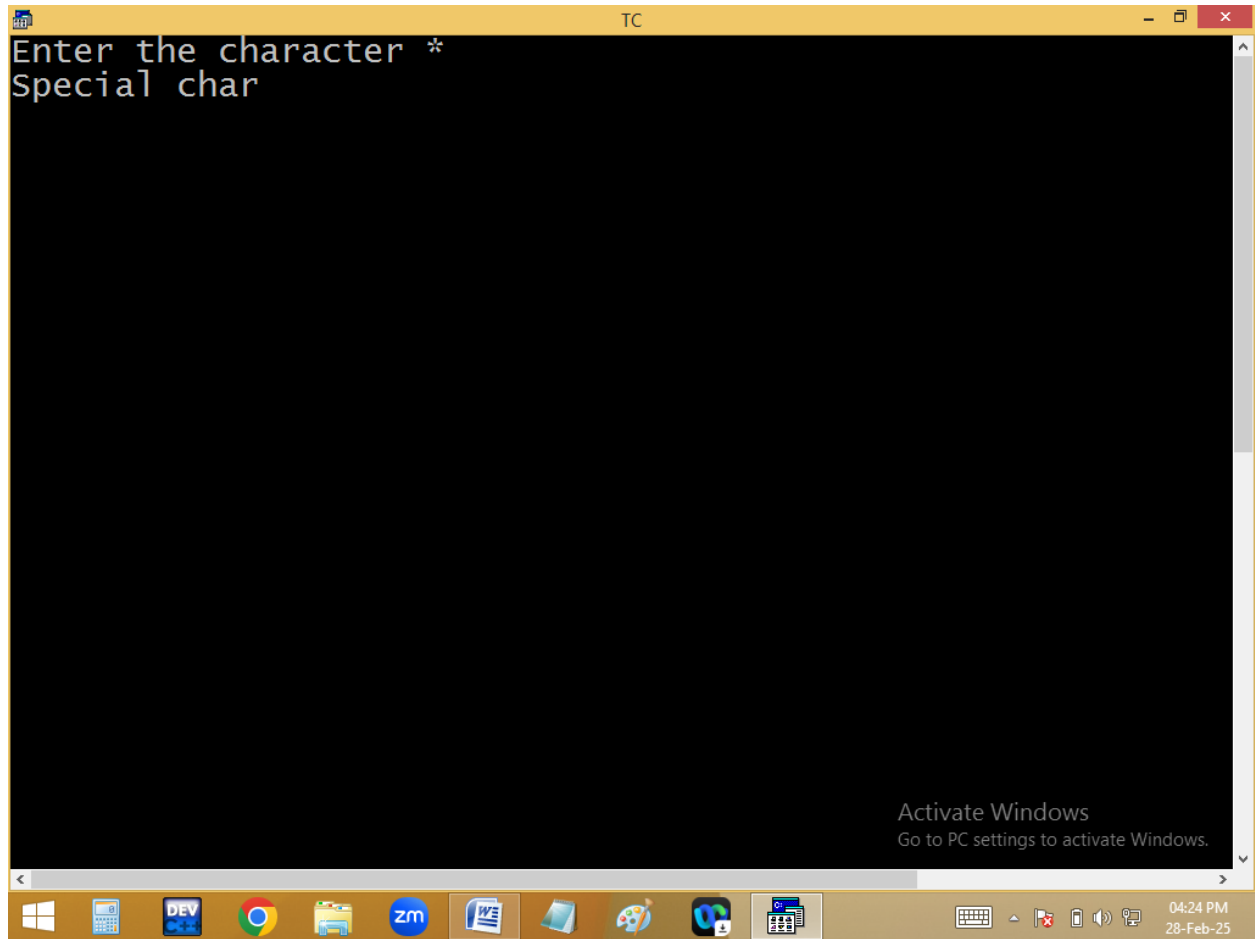
```
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); ch=getchar();
if(islower(ch))puts("Lower char");
else if(isupper(ch))puts("Upper case char");
else if(isdigit(ch))puts("Digit");
else if(isspace(ch))puts("Space");
else puts("Special char");
getch();
}
```

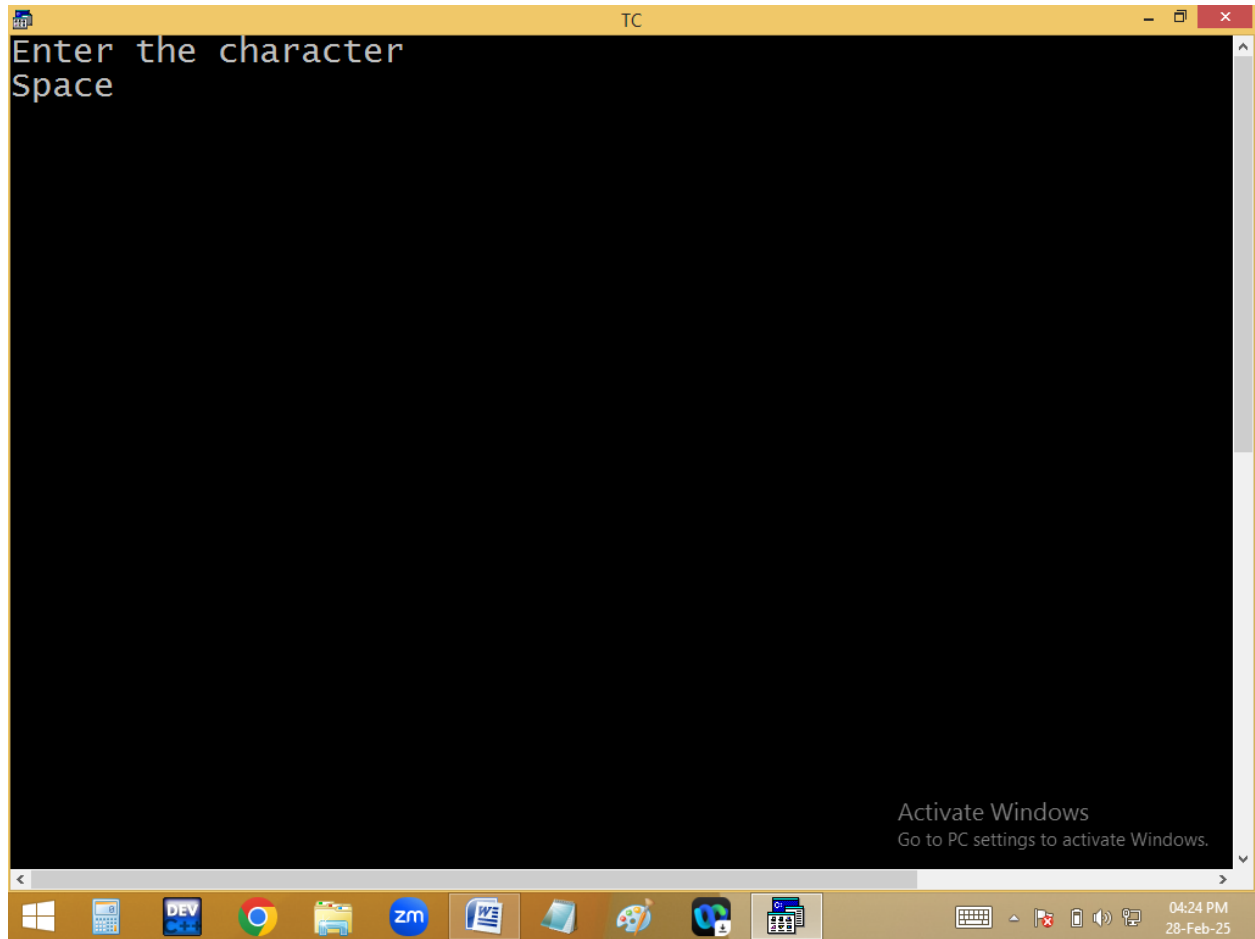
At the bottom of the window, there is a taskbar with various application icons including Windows, a calculator, a code editor, Google Chrome, a file explorer, a Zoom icon, a Word document, a folder, a paint application, a game, and a calculator. On the right side of the taskbar, the system clock shows "04:23 PM" and the date "28-Feb-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window, with the text "Go to PC settings to activate Windows."

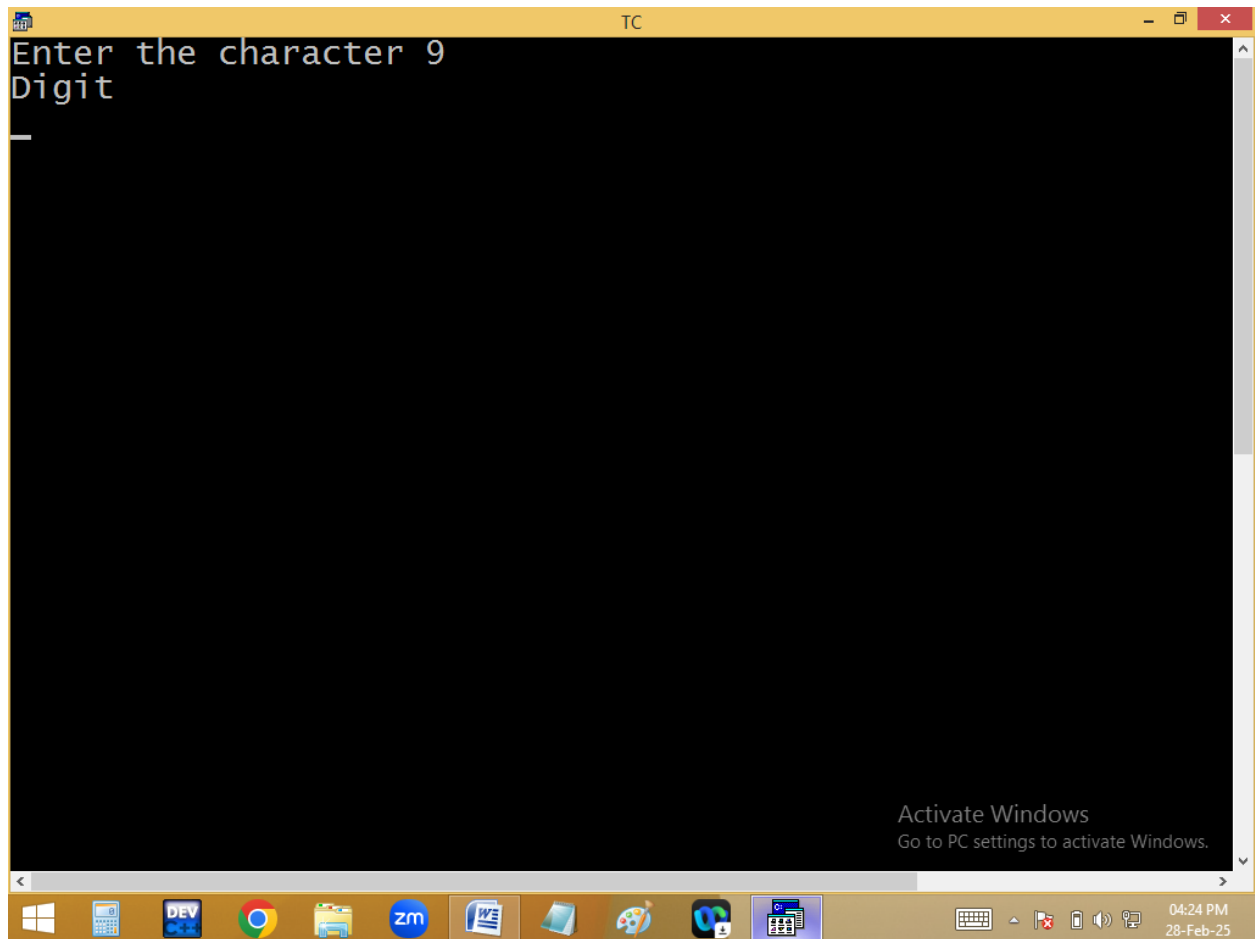






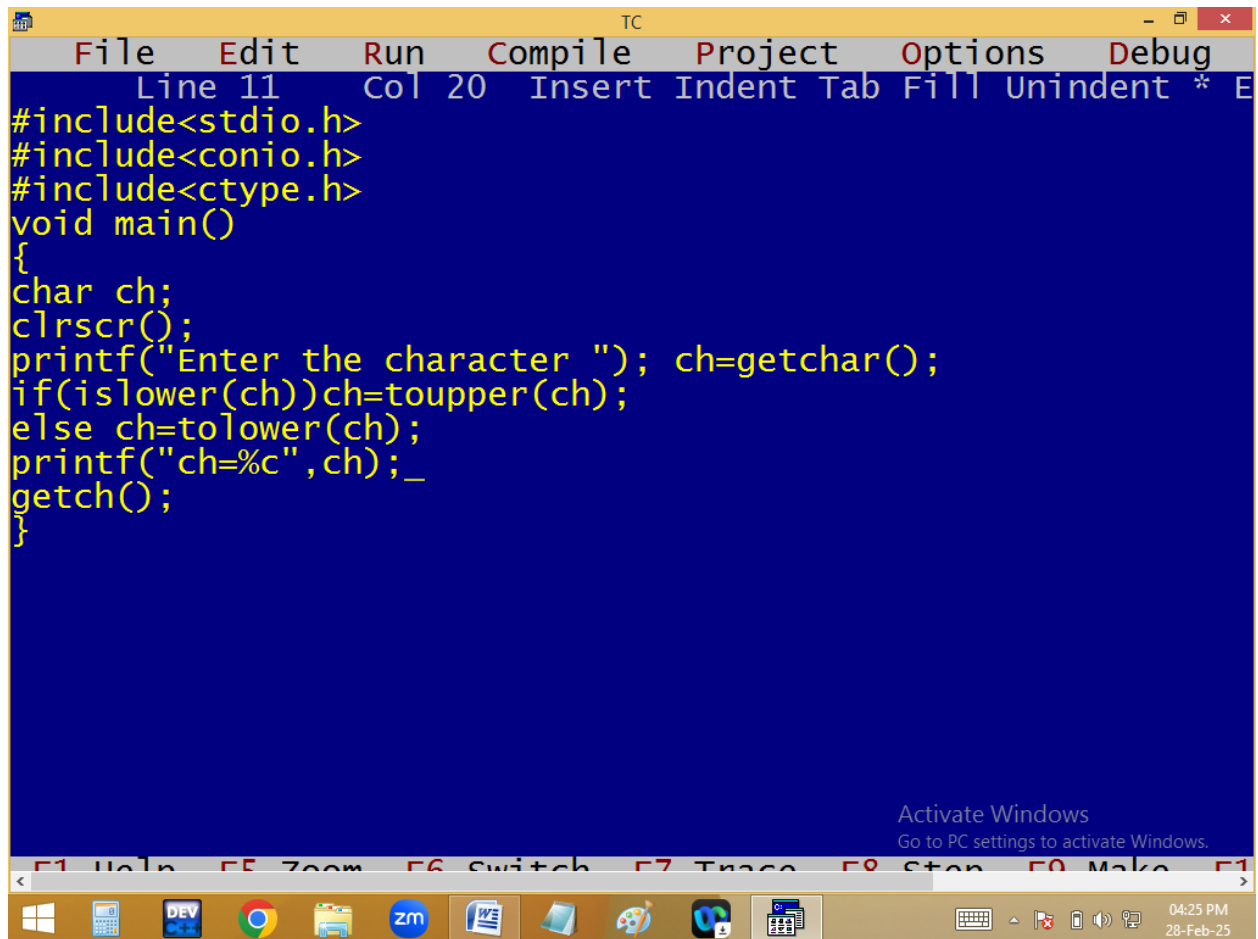






**Lower to upper / upper to lower:**

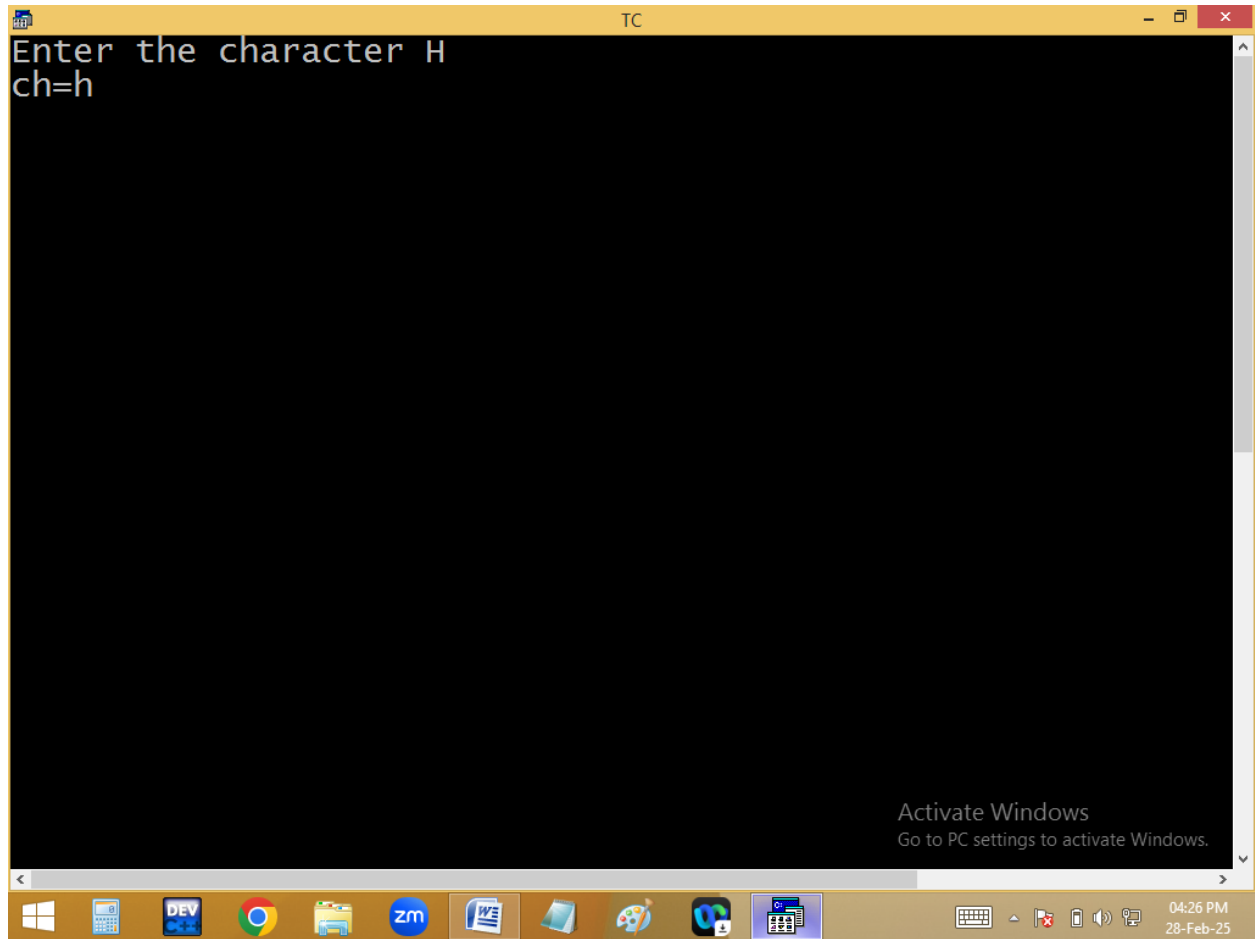
**Using predefined functions:**

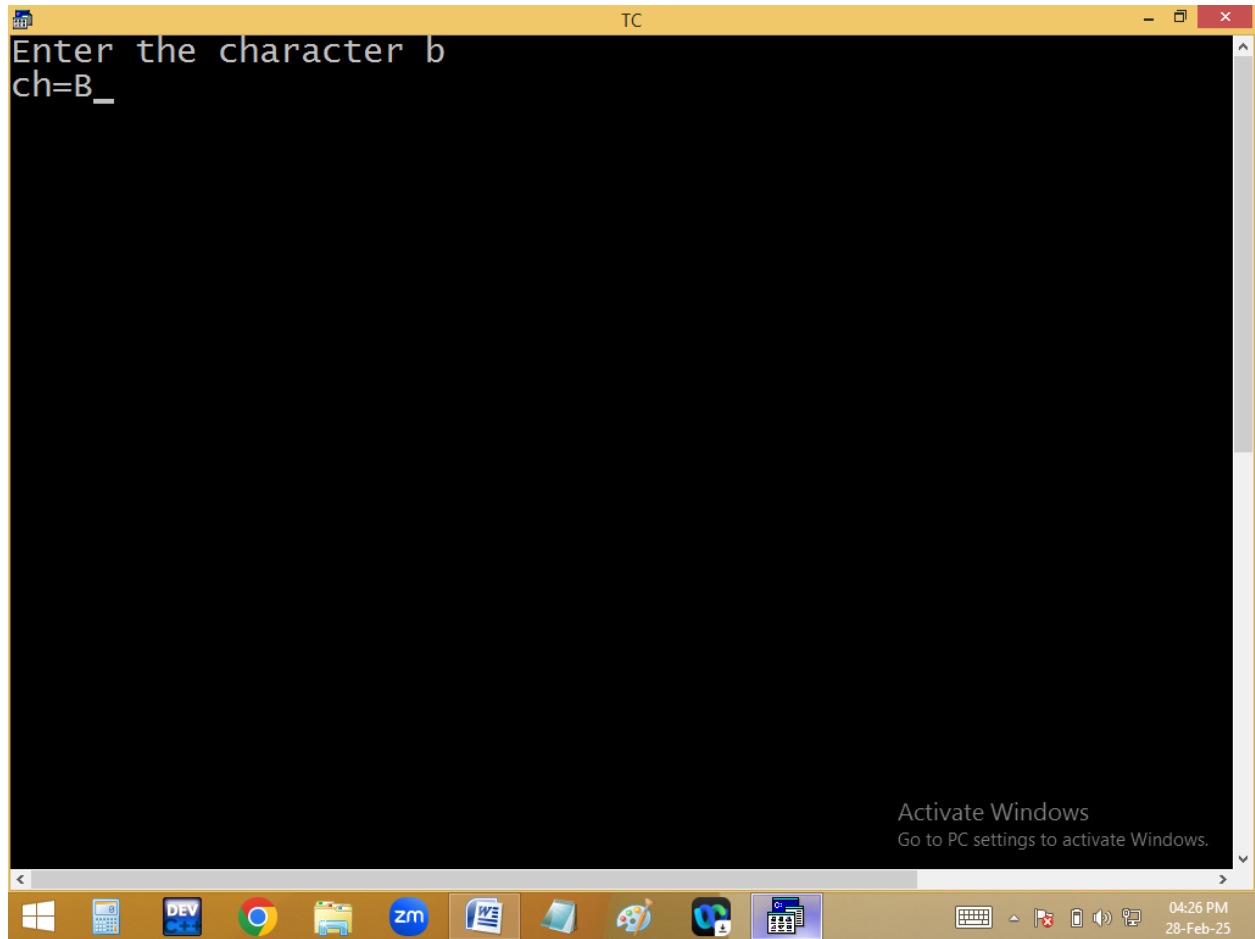


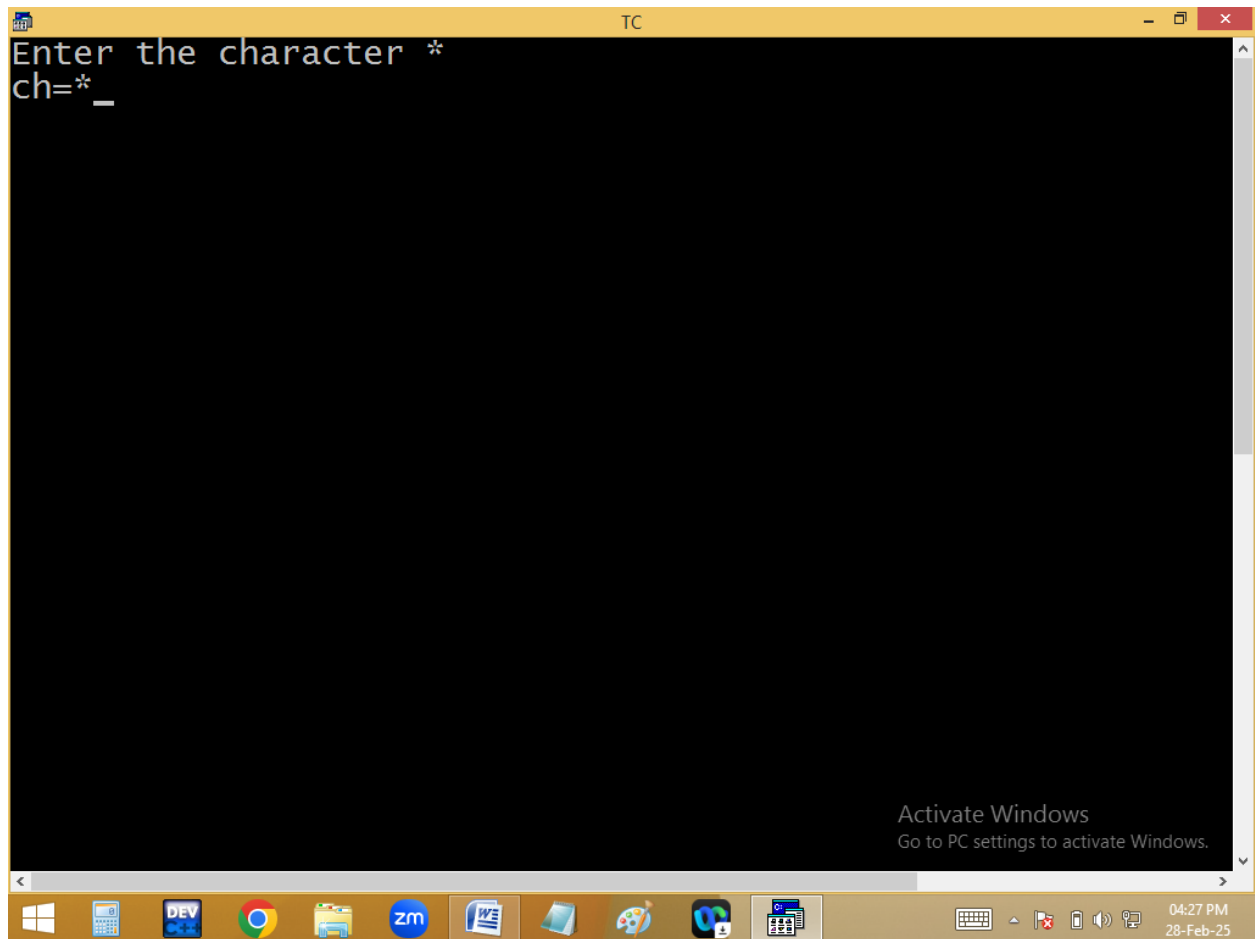
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar at the top reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status line indicates "Line 11 Col 20" and provides keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", and a wildcard "\*" followed by "E". The main editing area has a dark blue background with yellow text. It contains the following C code:

```
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); ch=getchar();
if(islower(ch))ch=toupper(ch);
else ch=tolower(ch);
printf("ch=%c",ch);_
getch();
}
```

At the bottom of the window, there is a taskbar with various application icons including Windows Explorer, Dev C++, Google Chrome, and Zoom. The system tray on the right shows the time as "04:25 PM" and the date as "28-Feb-25". An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.







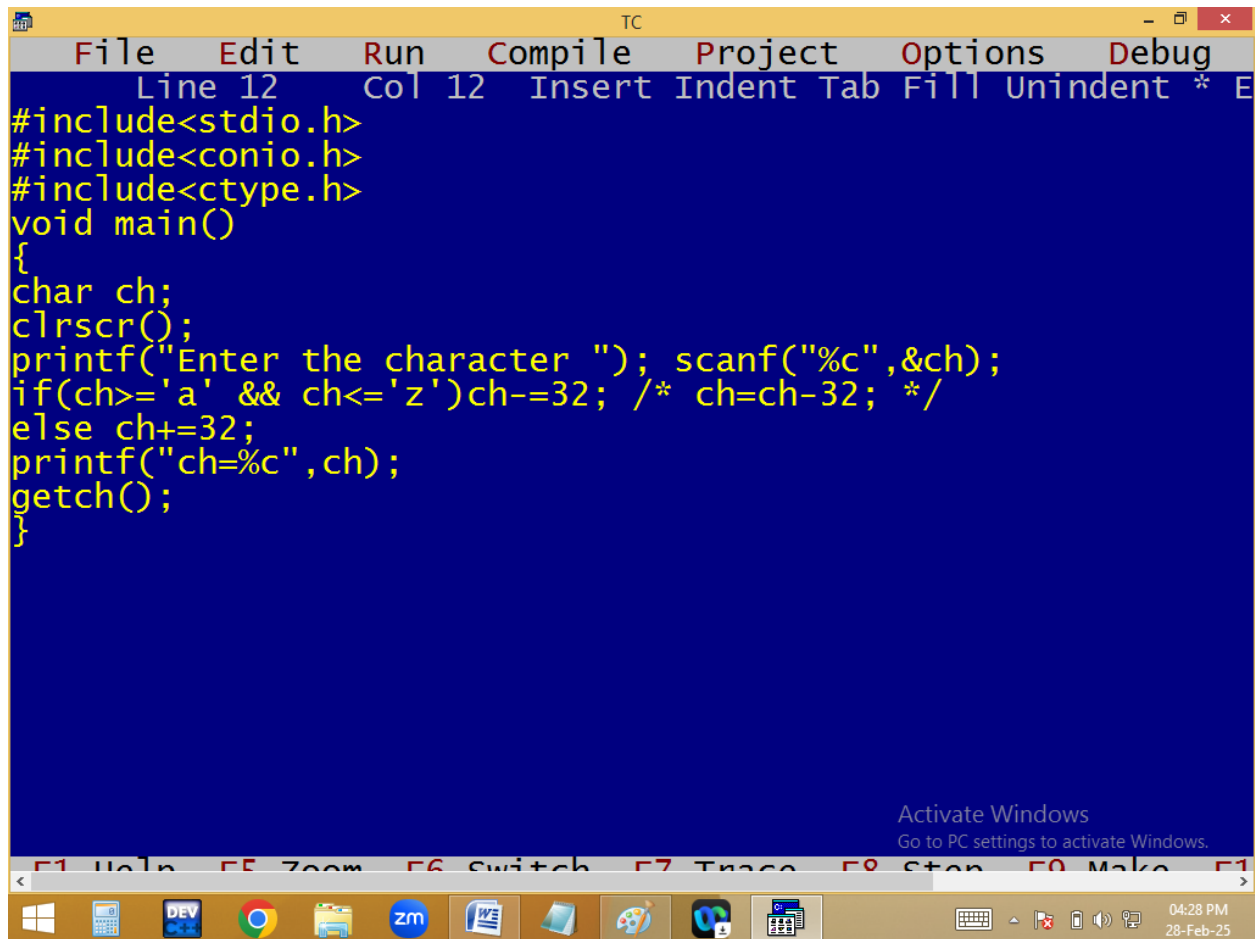
**Without using predefined functions:**

$$\begin{array}{r} 97 \\ - 32 \\ \hline \end{array}$$

$$A = 65$$

$$\begin{array}{r} 65 \\ + 32 \\ \hline 97 \end{array}$$

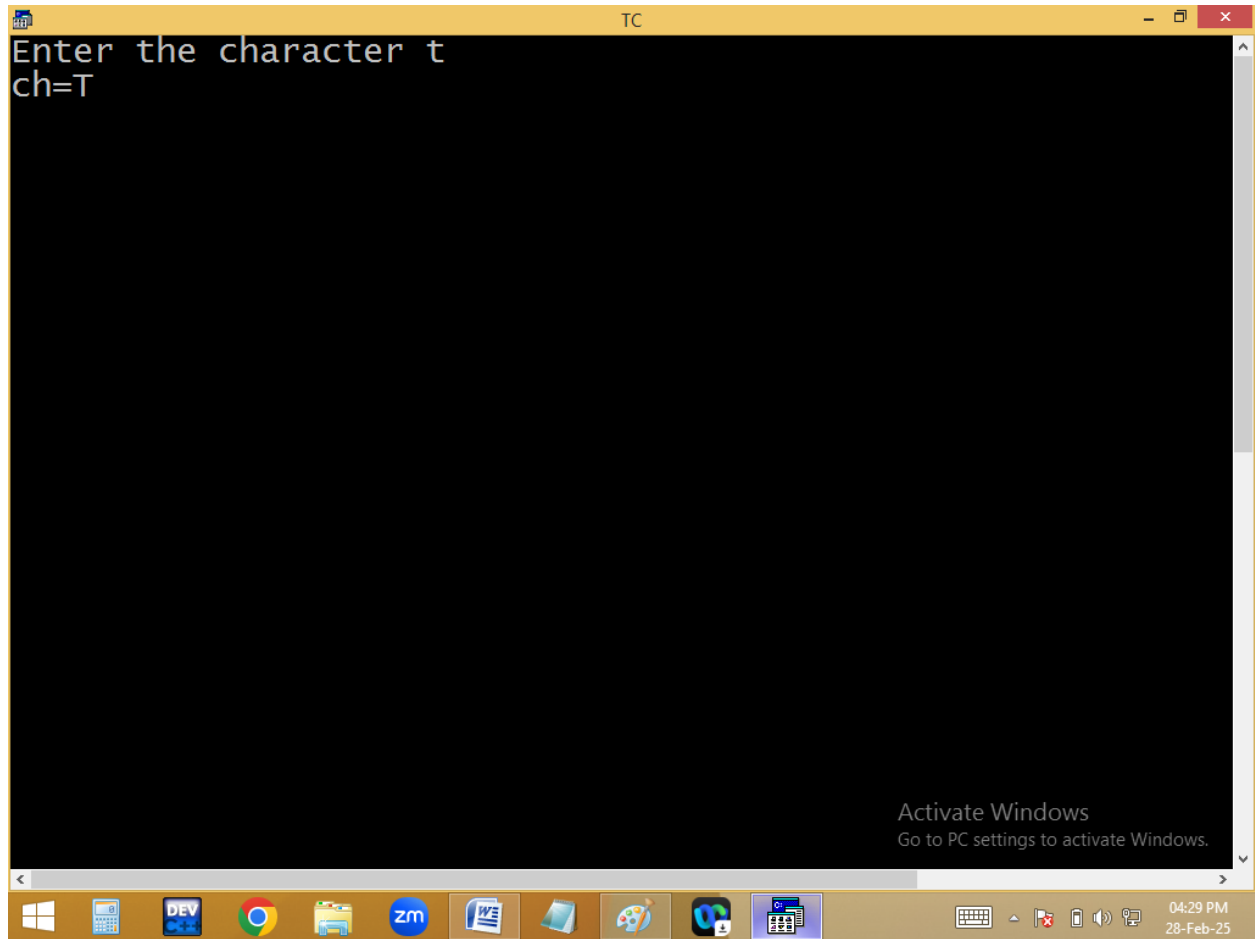


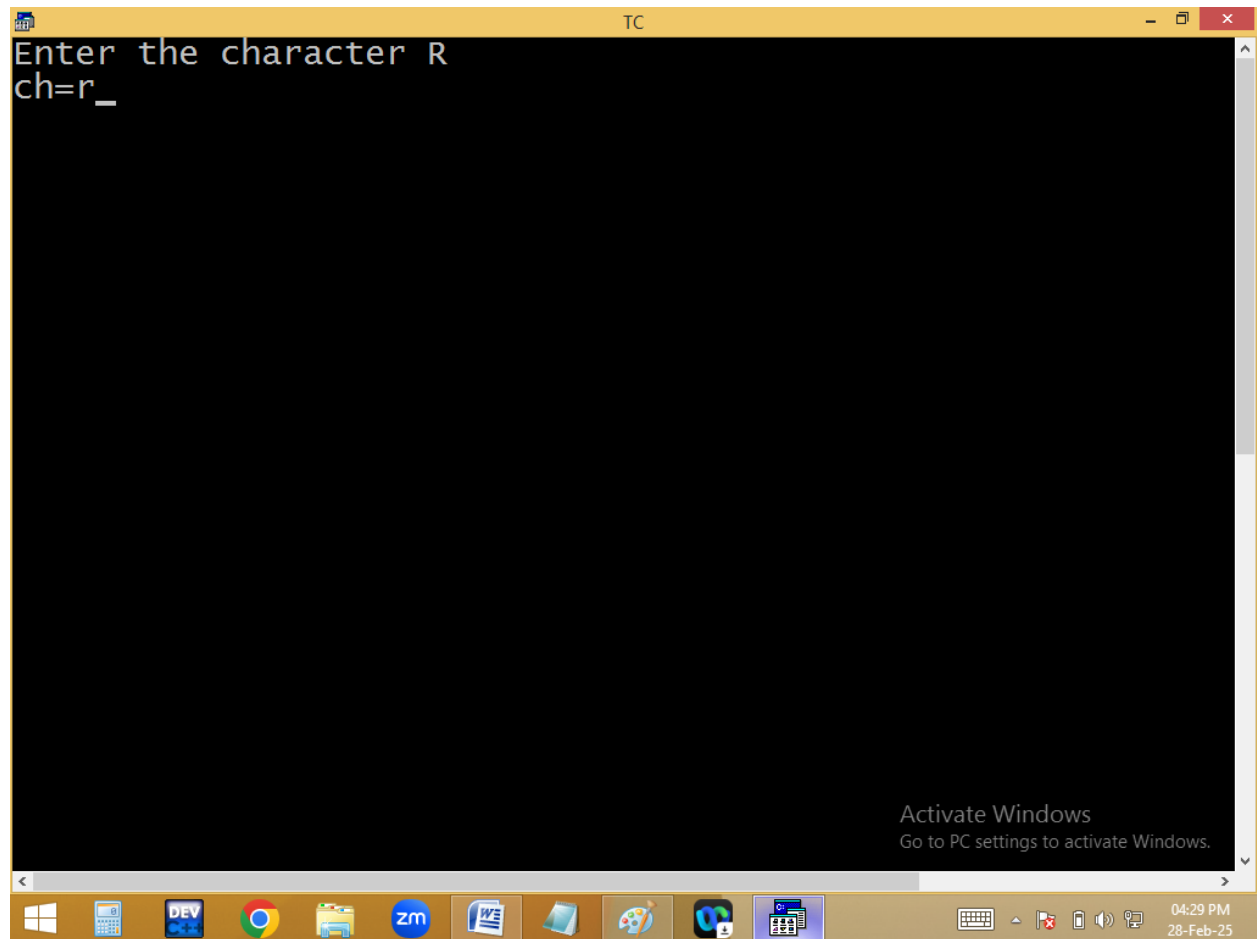


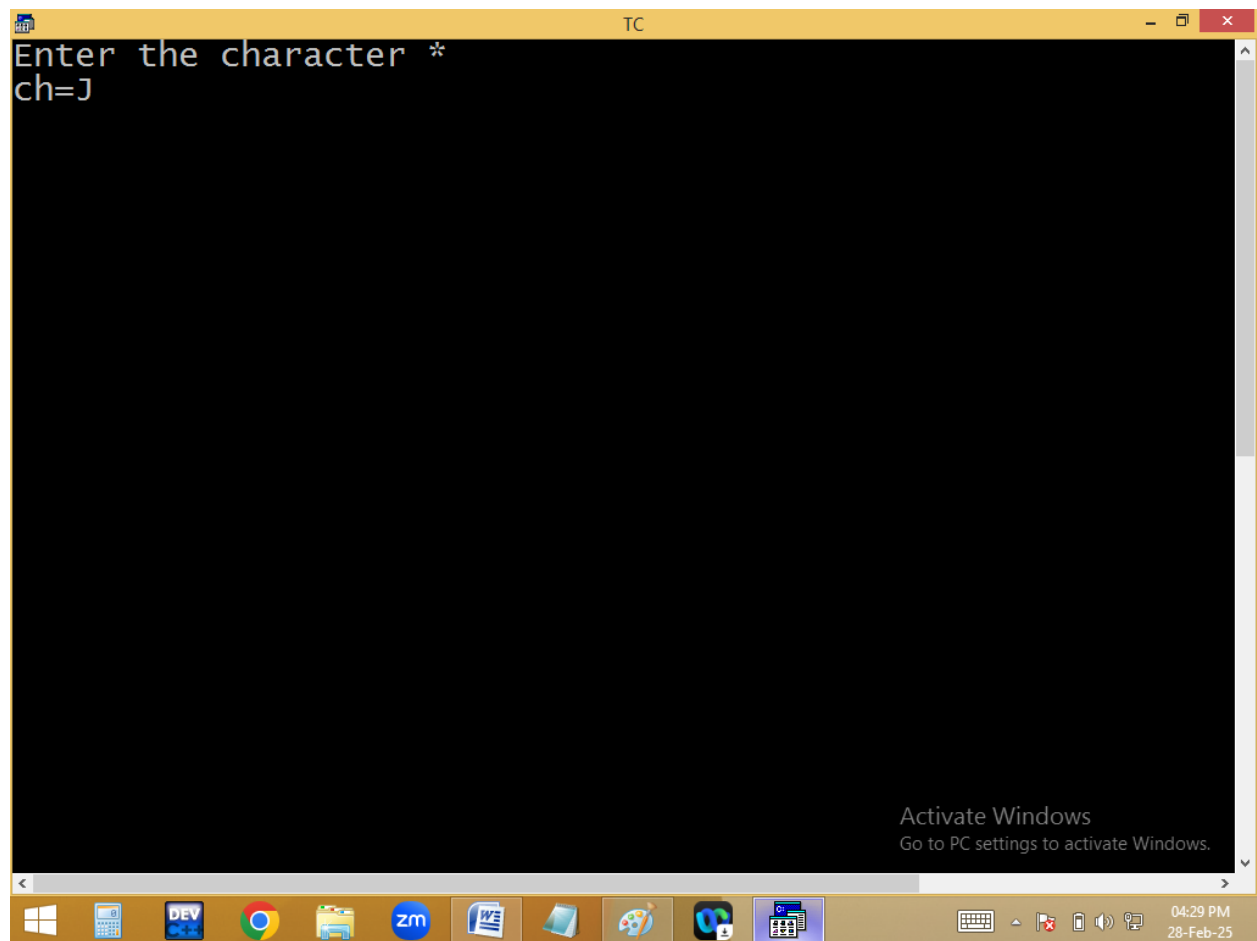
The image shows a screenshot of a Turbo C++ (TC) IDE window. The window has a yellow title bar with the text "TC" and standard window controls. The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". Below the menu bar, a status bar shows "Line 12", "Col 12", and various editing options like "Insert", "Indent", "Tab", "Fill", "Unindent", and "E". The main editing area has a blue background and contains the following C code:

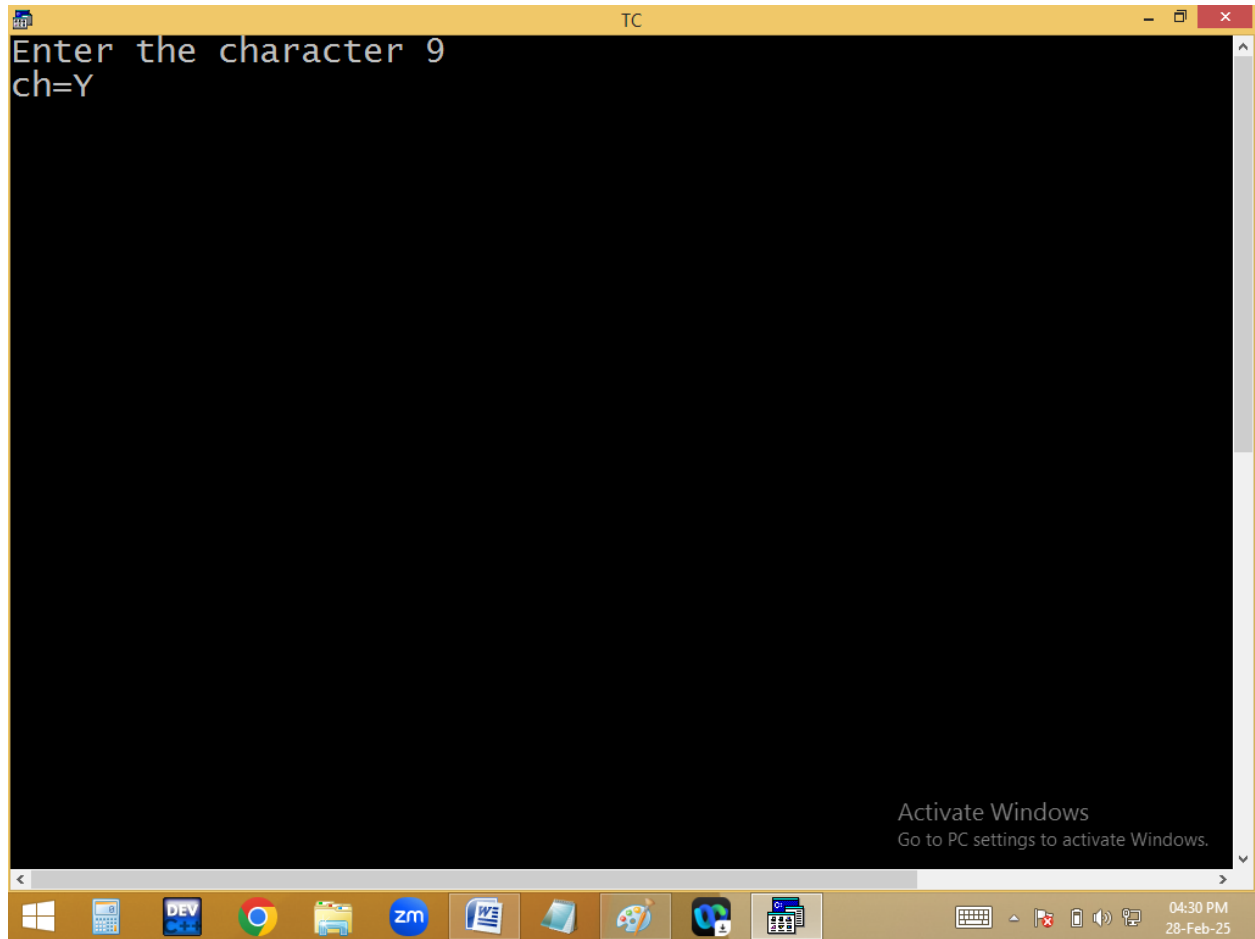
```
#include<stdio.h>
#include<conio.h>
#include<ctype.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>='a' && ch<='z')ch-=32; /* ch=ch-32; */
else ch+=32;
printf("ch=%c",ch);
getch();
}
```

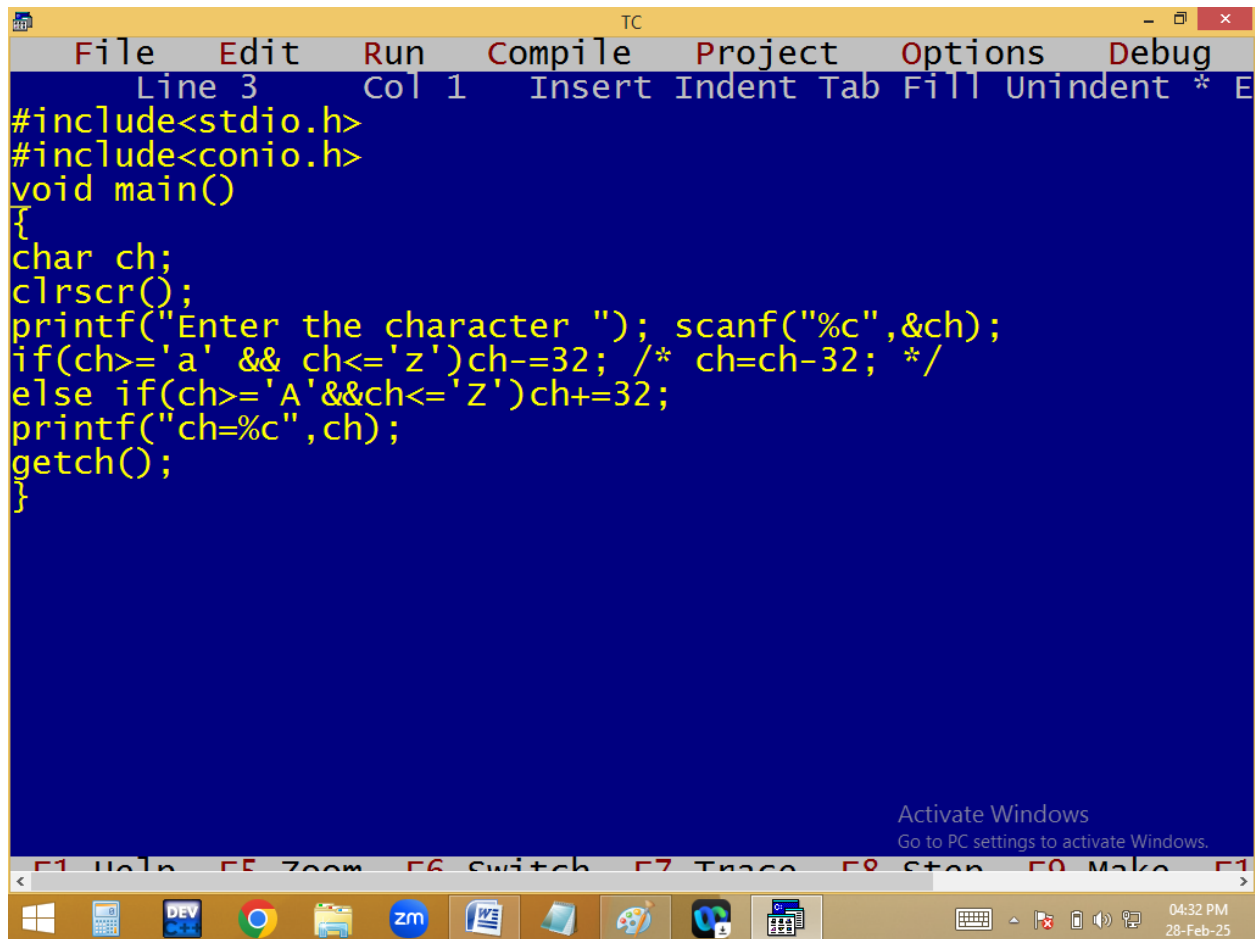
At the bottom of the window, there is a taskbar with various application icons including Windows, Calculator, DEV C++, Google Chrome, File Explorer, Zoom, Word, and others. A system tray on the right shows the time as 04:28 PM and the date as 28-Feb-25. An "Activate Windows" watermark is visible in the bottom right corner of the IDE window.







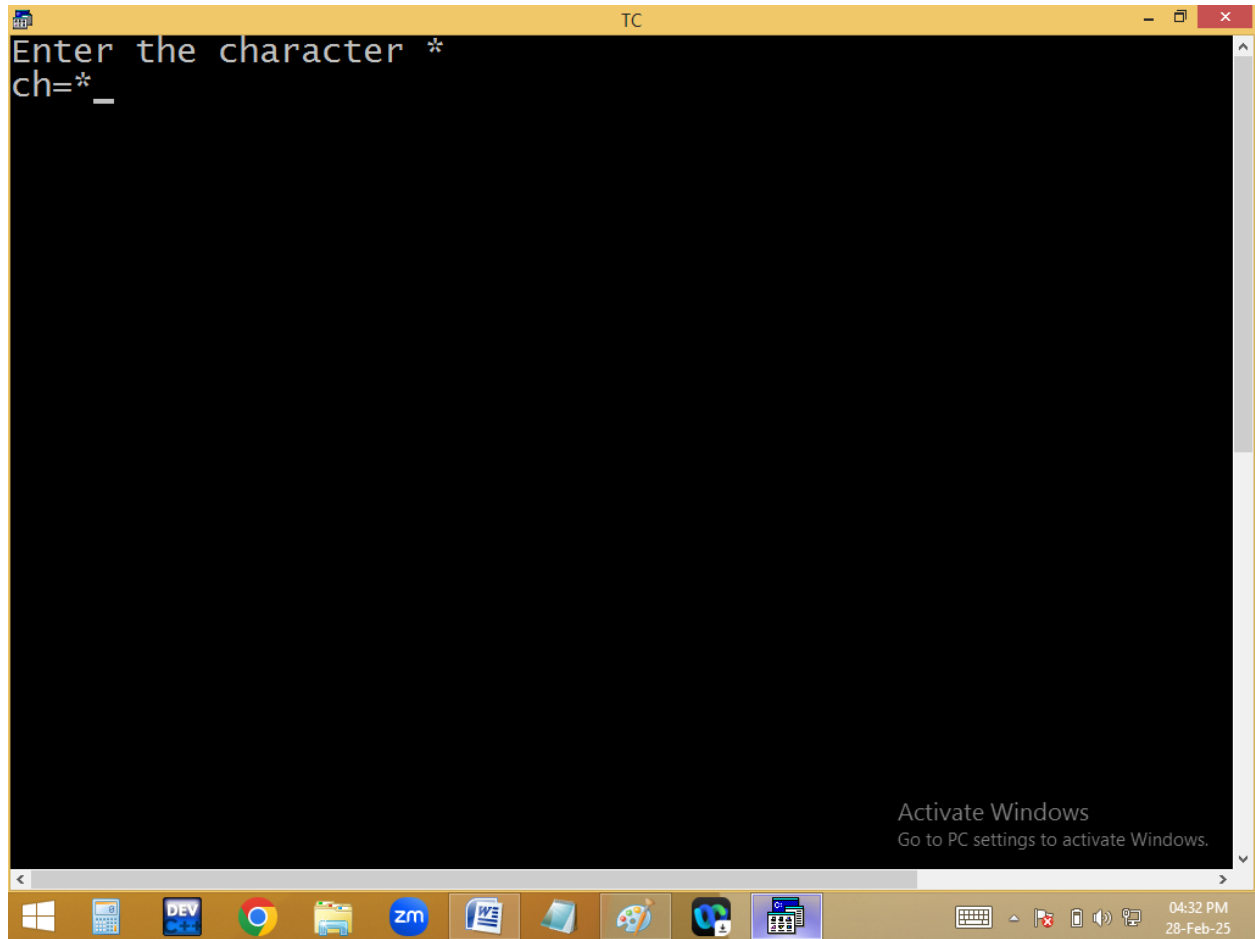


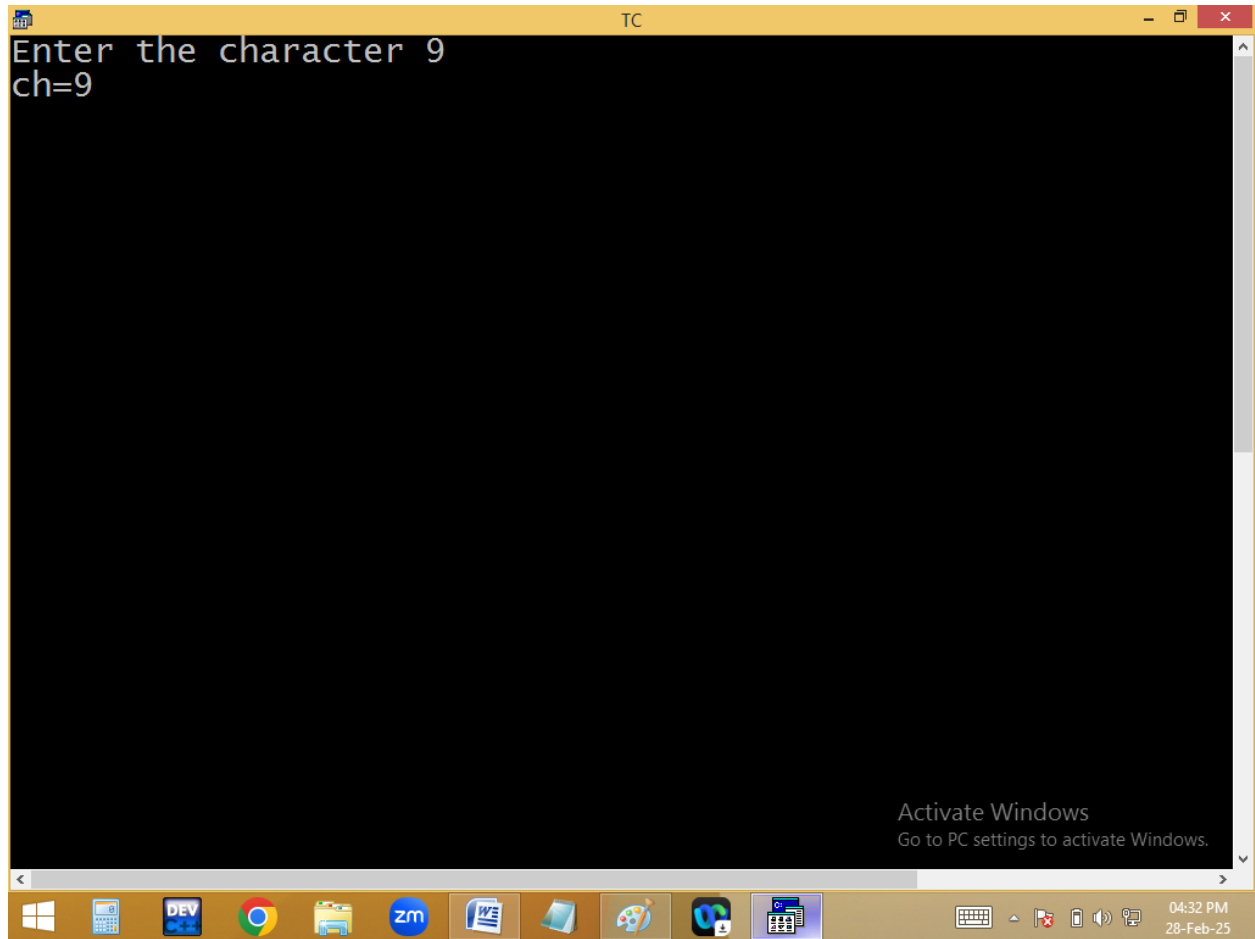


The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes File, Edit, Run, Compile, Project, Options, and Debug. The status bar at the top indicates "Line 3" and "Col 1". The main editing area has a blue background and contains the following C code:

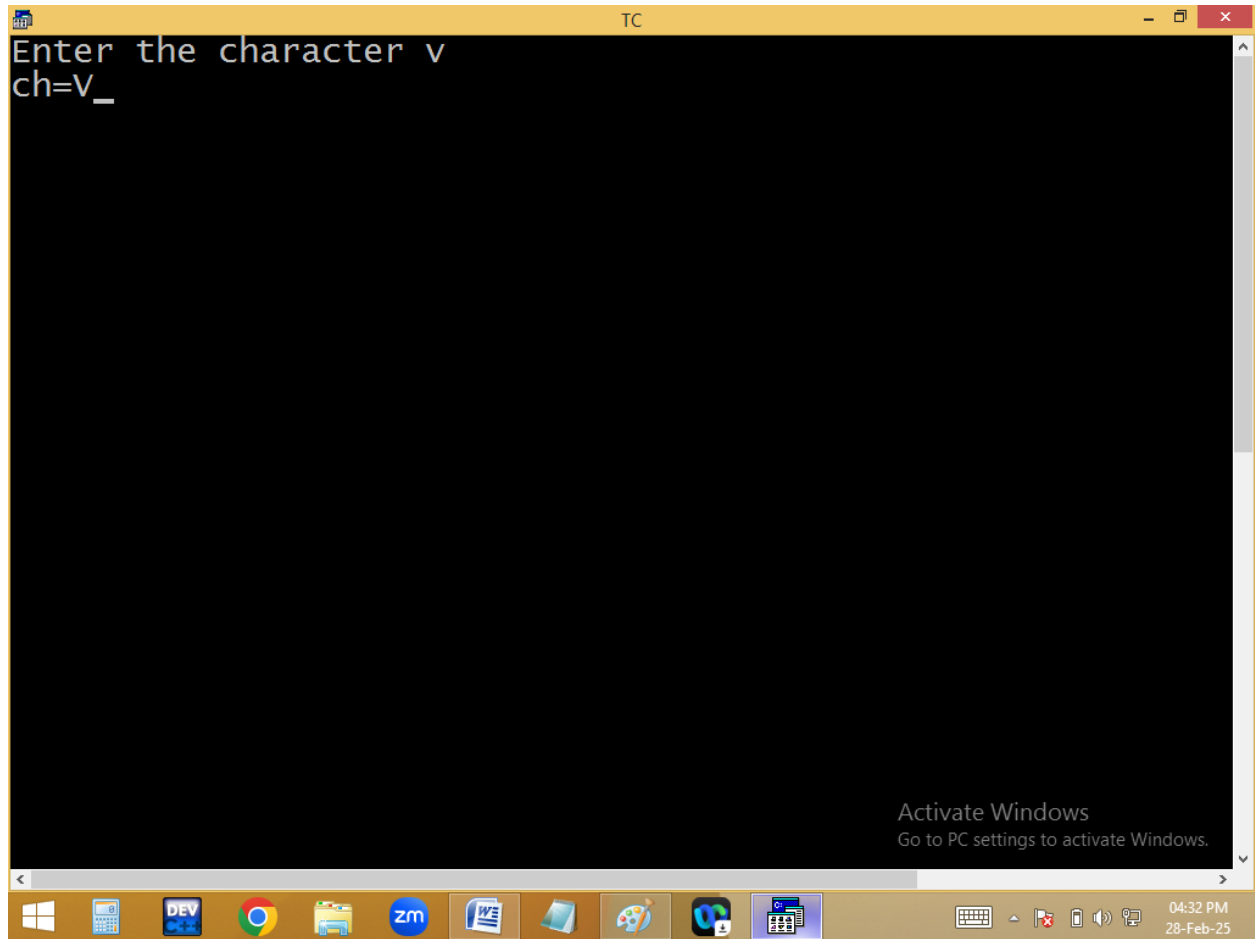
```
#include<stdio.h>
#include<conio.h>
void main()
{
char ch;
clrscr();
printf("Enter the character "); scanf("%c",&ch);
if(ch>='a' && ch<='z')ch-=32; /* ch=ch-32; */
else if(ch>='A'&&ch<='Z')ch+=32;
printf("ch=%c",ch);
getch();
}
```

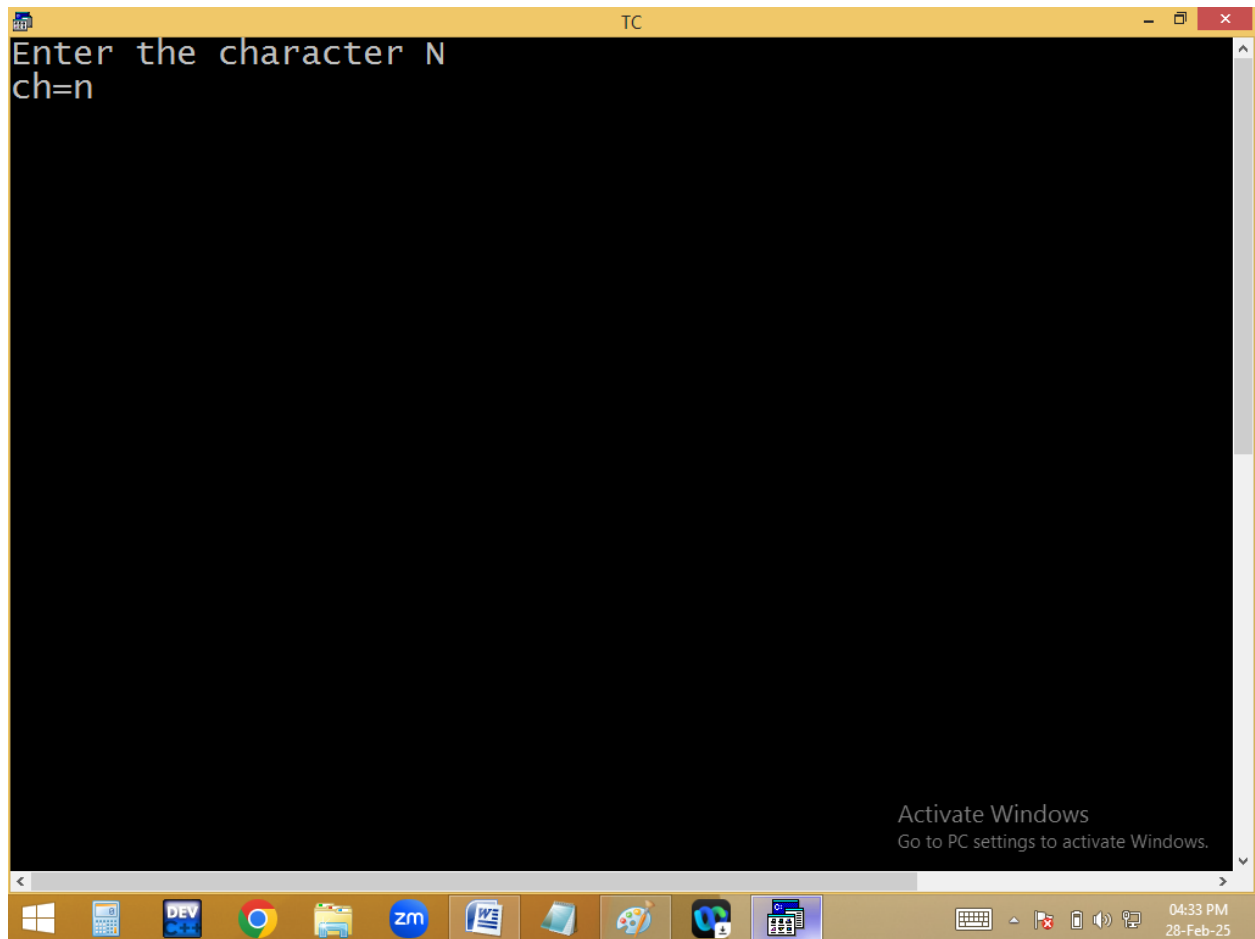
At the bottom of the IDE window, there is a text prompt: "Activate Windows Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for Windows, Task View, File Explorer, Google Chrome, Dev C++, Zoom, and other applications. The system tray on the right shows the date and time as "04:32 PM 28-Feb-25".











**Electricity bill generation:**

### **DOMESTIC SLAB SYSTEM**

| <b>UNITS</b>   | <b>UNIT PRICE</b> |
|----------------|-------------------|
| <b>1-50</b>    | <b>1.45</b>       |
| <b>51-100</b>  | <b>2.80</b>       |
| <b>101-200</b> | <b>3.05</b>       |
| <b>201-300</b> | <b>4.75</b>       |
| <b>301-500</b> | <b>6.00</b>       |
| <b>&gt;500</b> | <b>6.25</b>       |

$$51U = 50 * 1.45 + 1 * 2.8 =$$

$$30\text{UNITS} * 1.45=43.5$$

$$80\text{UNITS} = 50*1.45 + 30*2.8=156.5$$

$$580\text{U} = 50*1.45 + 50*2.8 + 100*3.05+100*4.75+200*6+80*6.25=2692.5$$

```
#include<stdio.h>
```

```
#include<conio.h>
```

```
void main()
```

```
{
```

```
long serno, pre, cur, units;
```

```
char name[30];
```

```
float amt;
```

```
clrscr();
```

```
printf("Enter serial no "); scanf("%ld",&serno);
```

```
fflush(stdin);
```

```
printf("Enter name "); gets(name);
```

```
printf("Enter previous month reading ");scanf("%ld",&pre);
```

```
current:
```

```
printf("Enter current month reading ");scanf("%ld",&cur);
```

```
if(cur<pre)
```

```
{puts("\aCheck current month reading");goto current;}
```

```
units = cur - pre;
```

```
if(units<=50) amt=units*1.45;
```

```
else if(units<=100)amt=50*1.45+(units-50)*2.8;
```

```
else if(units<=200)amt=50*1.45+50*2.8+(units-100)*3.05;
```

```
else if(units<=300)amt=50*1.45+50*2.8+100*3.05+(units-  
200)*4.75;
```

```
else
```

```
if(units<=500)amt=50*1.45+50*2.8+100*3.05+100*4.75+(units  
-300)*6;
```

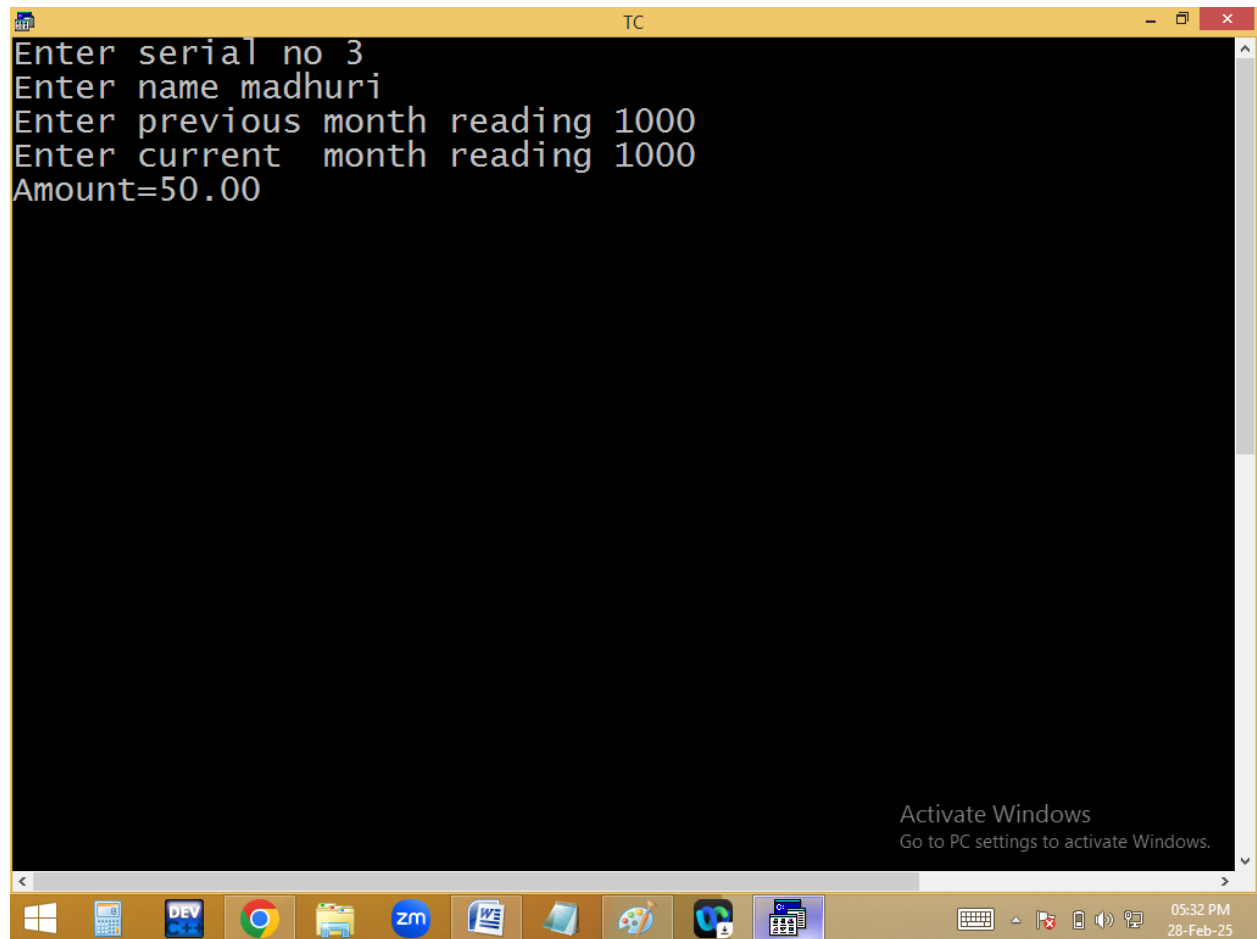
```
else amt=50*1.45+50*2.8+100*3.05+100*4.75+200*6+(units-  
500)*6.25;
```

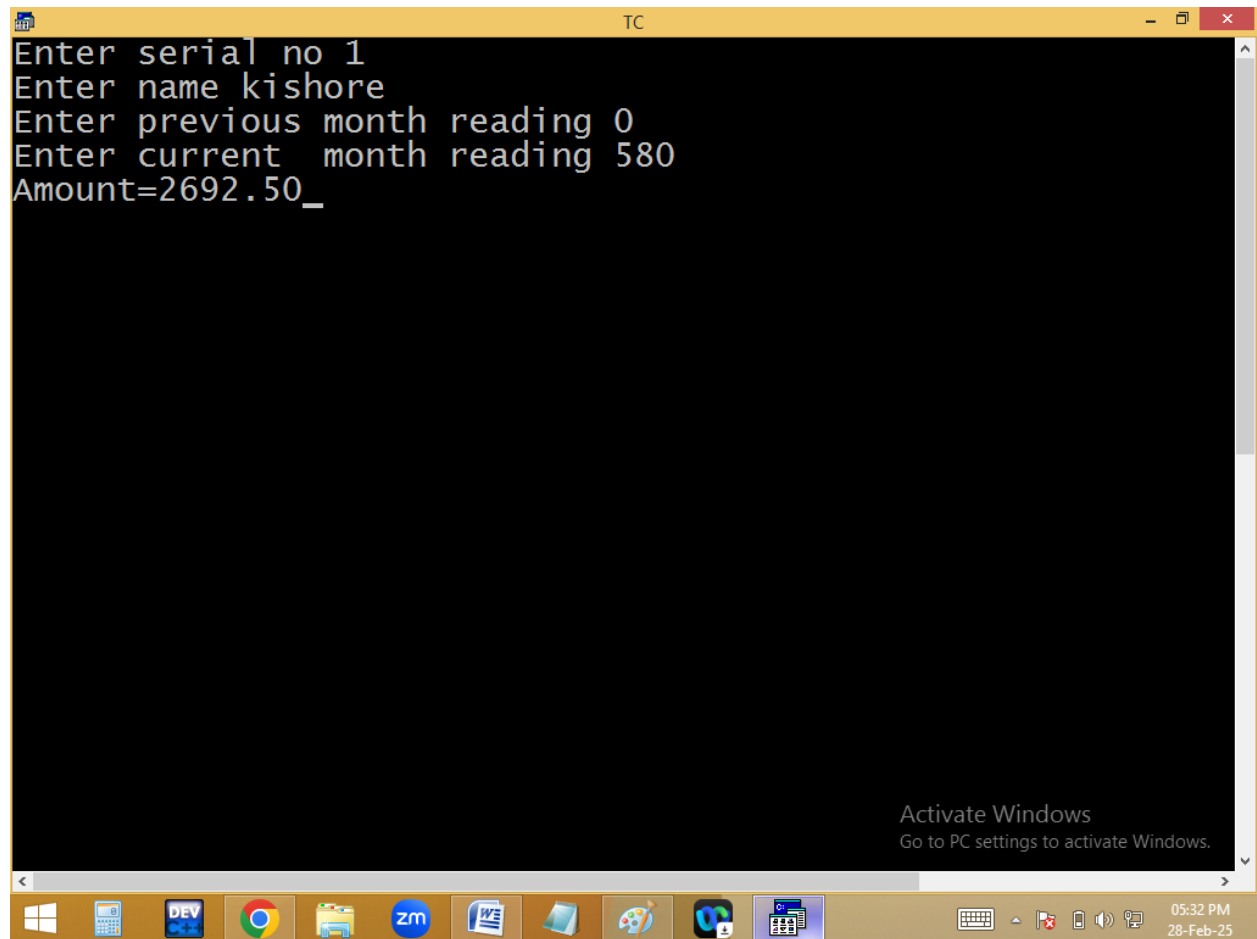
```
if(amt<50)amt=50;
```

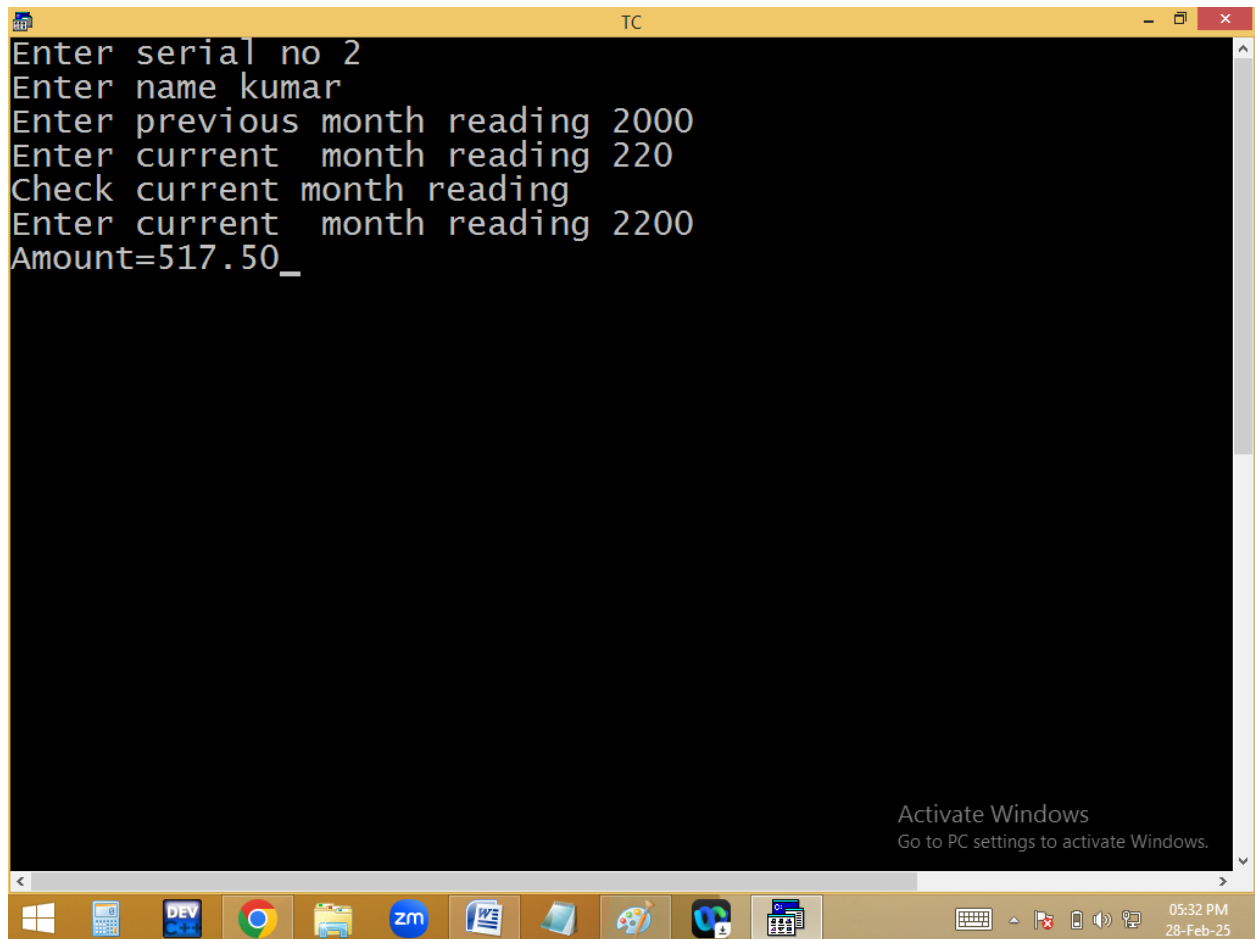
```
printf("Amount=%.2f",amt);
```

```
getch();
```

}







```
Enter serial no 2
Enter name kumar
Enter previous month reading 2000
Enter current month reading 220
Check current month reading
Enter current month reading 2200
Amount=517.50_
```

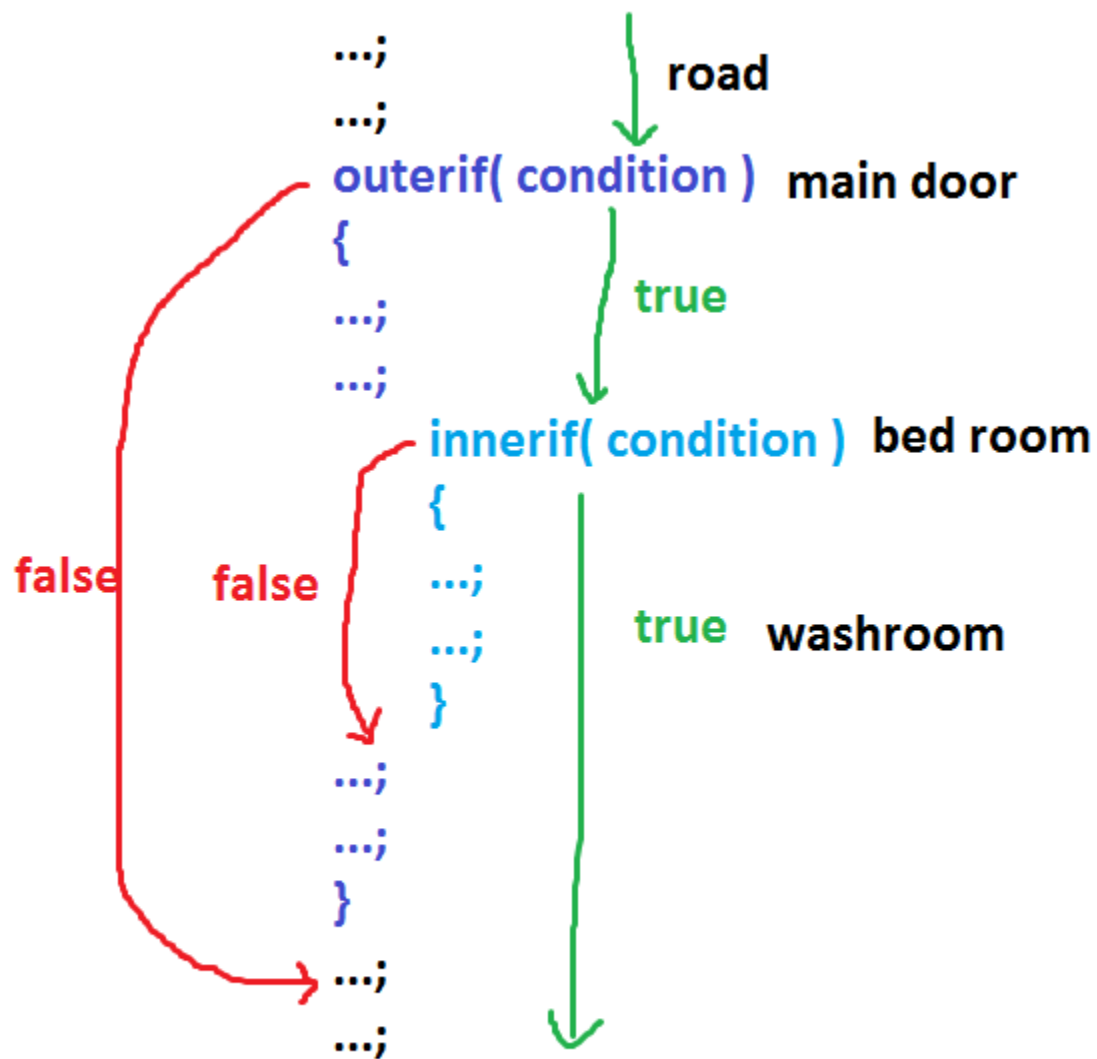
Activate Windows  
Go to PC settings to activate Windows.

05:32 PM  
28-Feb-25

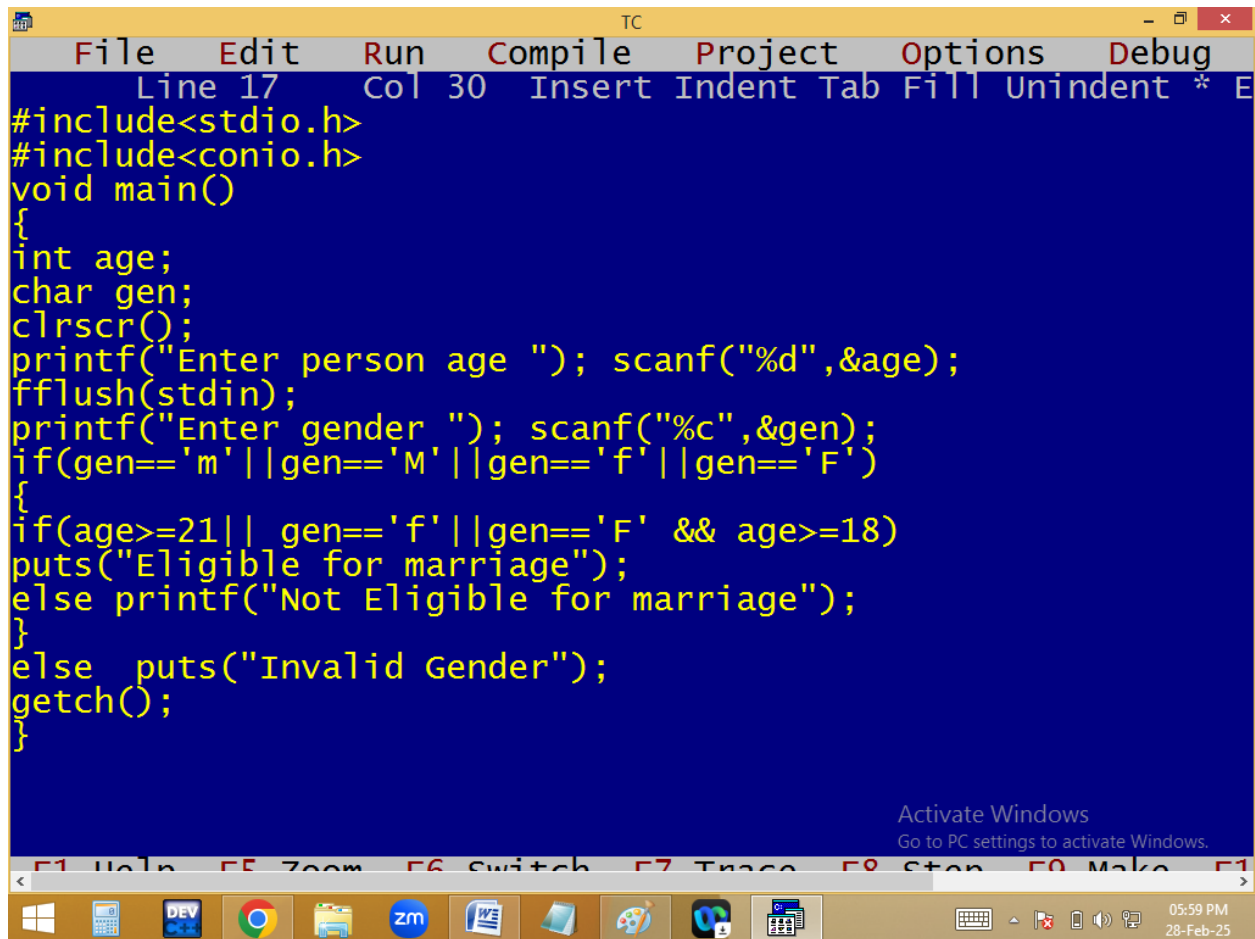
## Nested if:

If within if is called nested if.





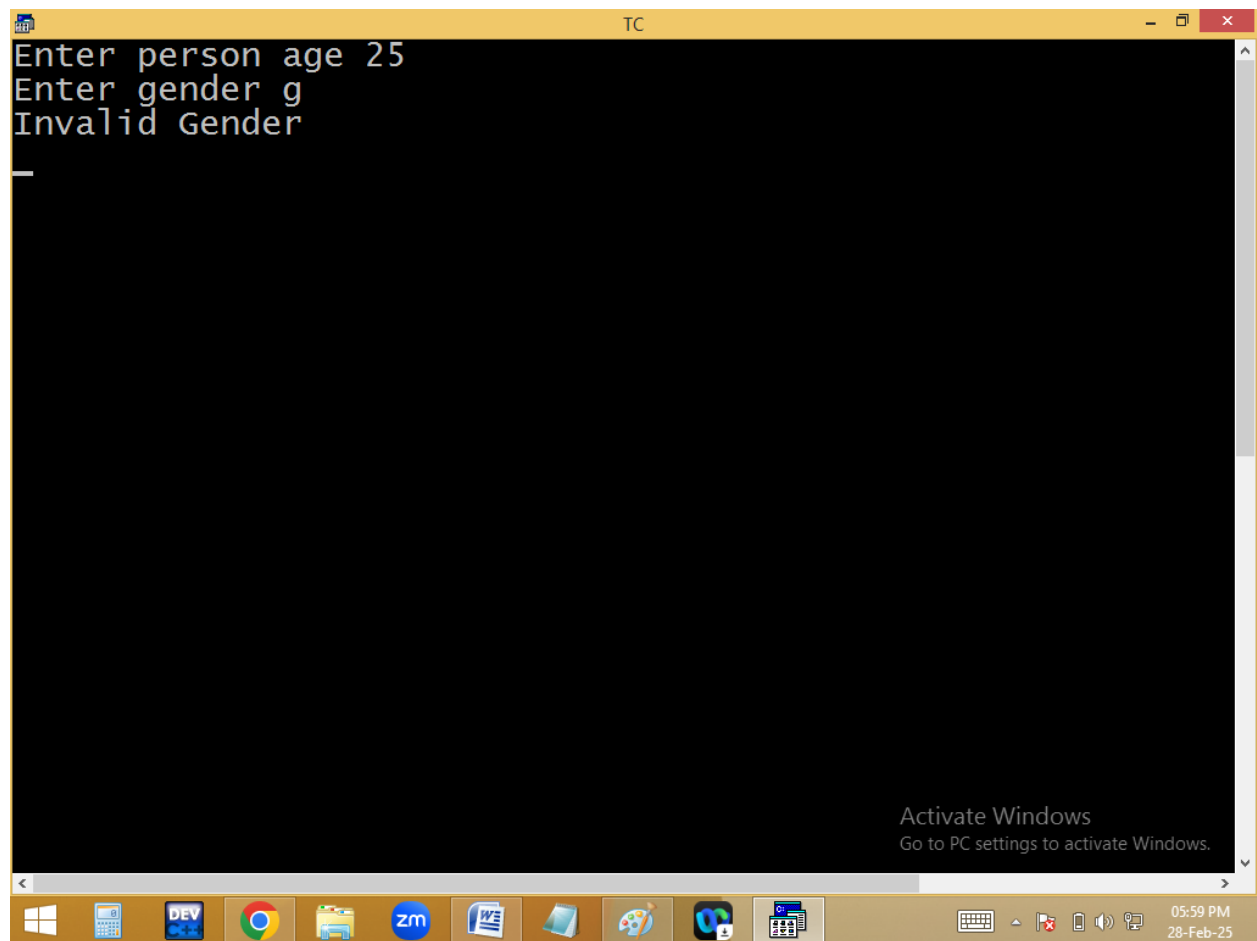
Read a person age and gender. Find that person is eligible for marriage or not?

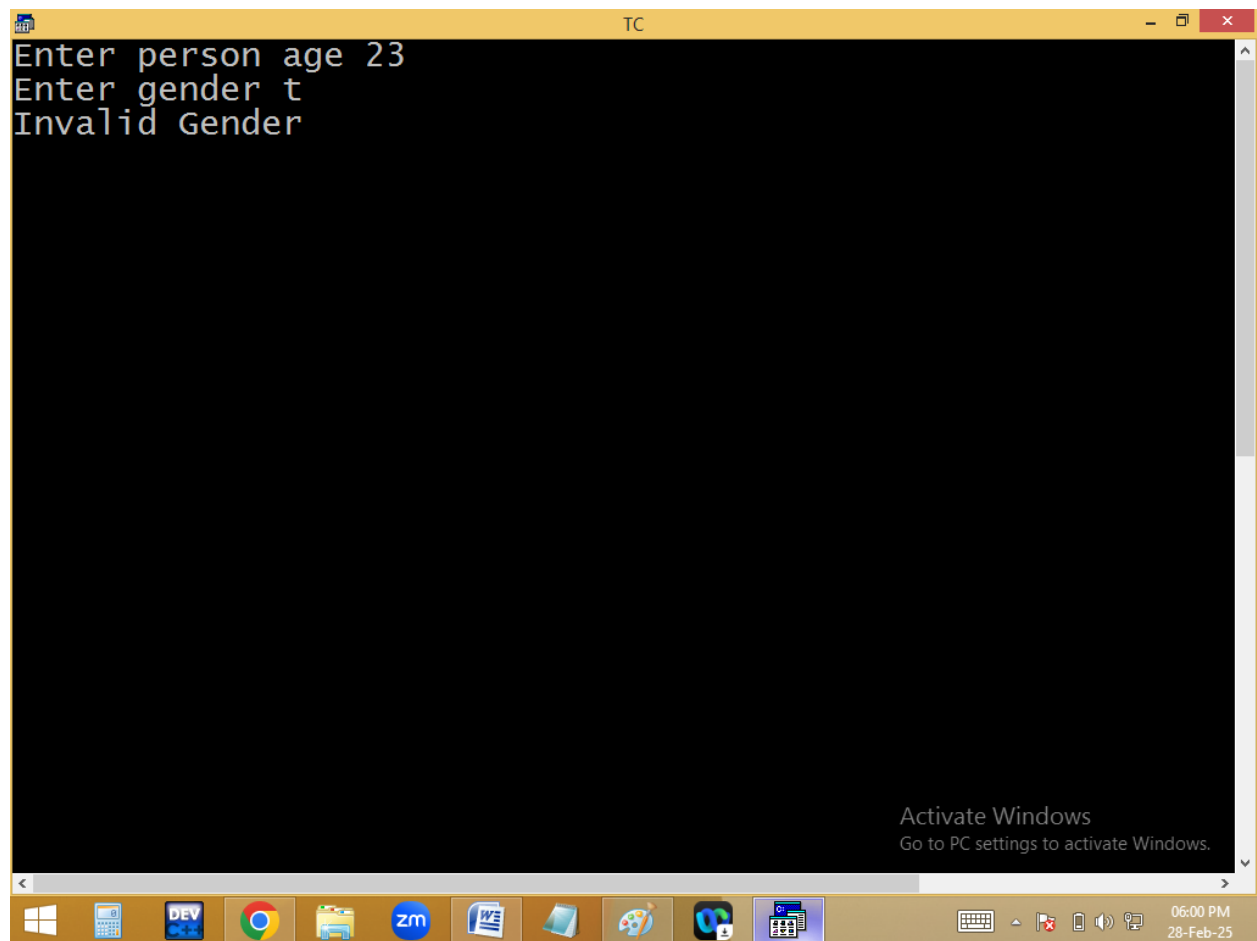


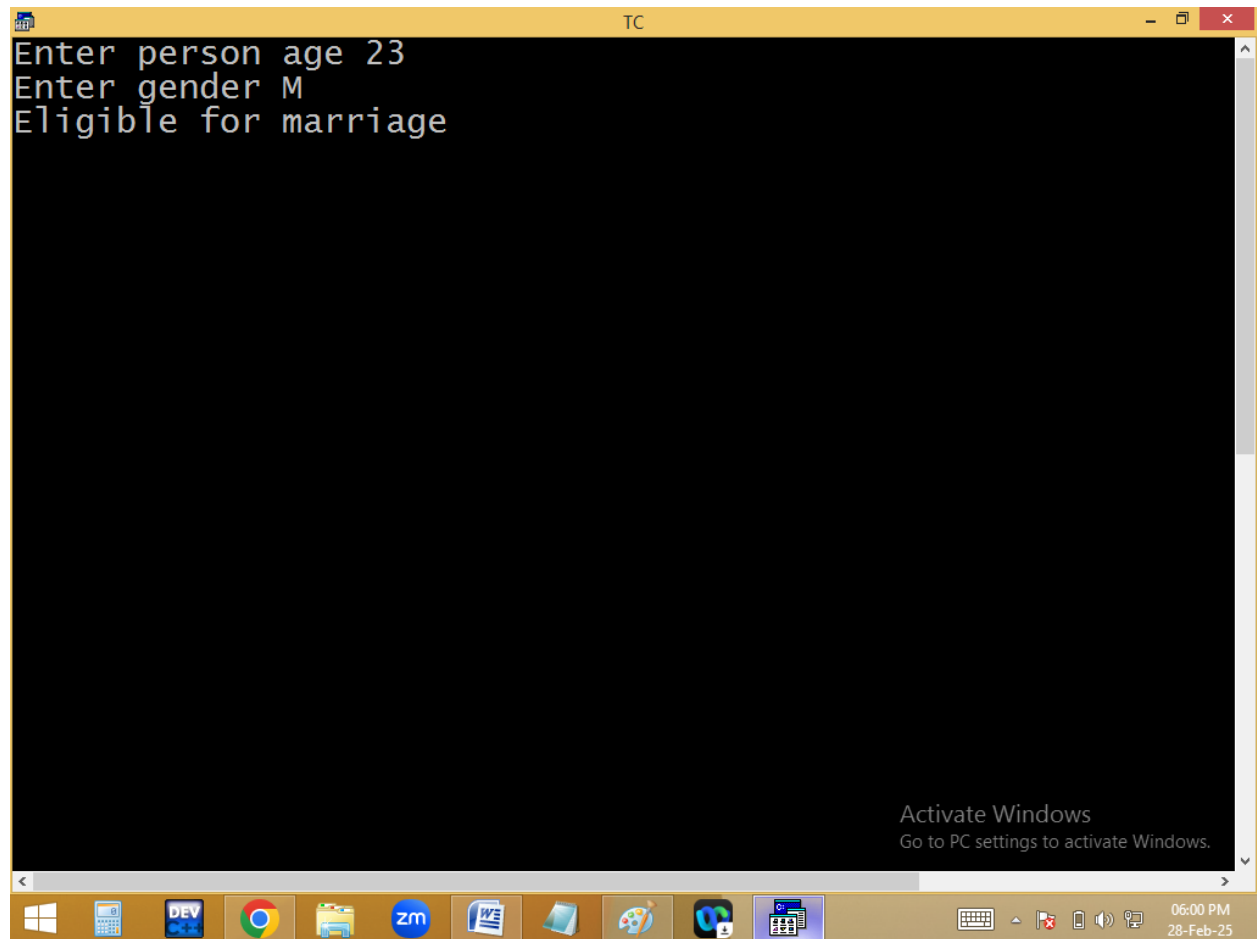
The image shows a screenshot of a Turbo C++ (TC) IDE window. The title bar reads "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 17", "Col 30", and lists keyboard shortcuts: "Insert", "Indent", "Tab", "Fill", "Unindent", "\*", and "E". The main editing area has a blue background with yellow text. It contains a C program that prompts the user for age and gender, then checks if they are eligible for marriage based on specific age and gender criteria. The code is as follows:

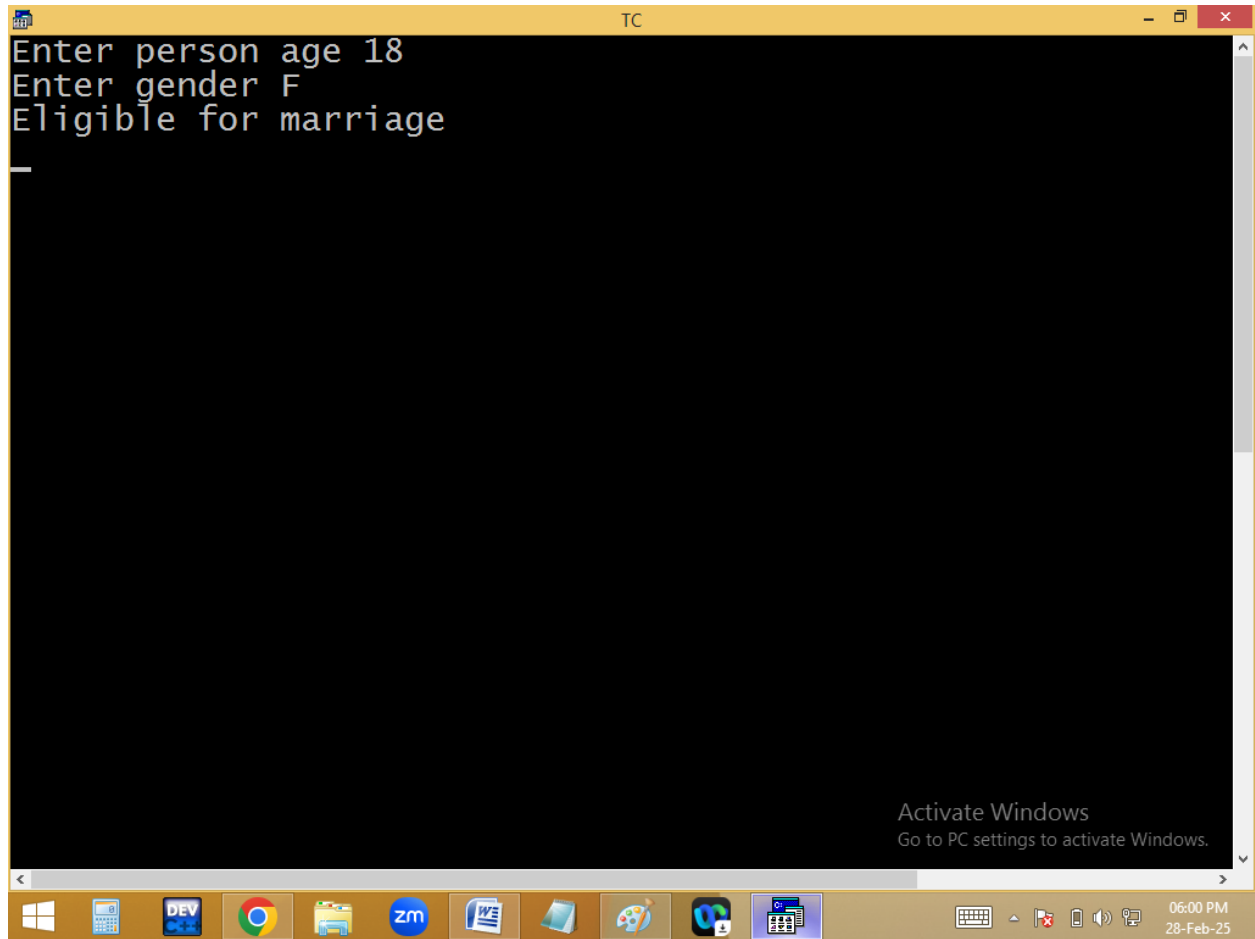
```
#include<stdio.h>
#include<conio.h>
void main()
{
    int age;
    char gen;
    clrscr();
    printf("Enter person age "); scanf("%d",&age);
    fflush(stdin);
    printf("Enter gender "); scanf("%c",&gen);
    if(gen=='m' || gen=='M' || gen=='f' || gen=='F')
    {
        if(age>=21 || gen=='f' || gen=='F' && age>=18)
            puts("Eligible for marriage");
        else printf("Not Eligible for marriage");
    }
    else puts("Invalid Gender");
    getch();
}
```

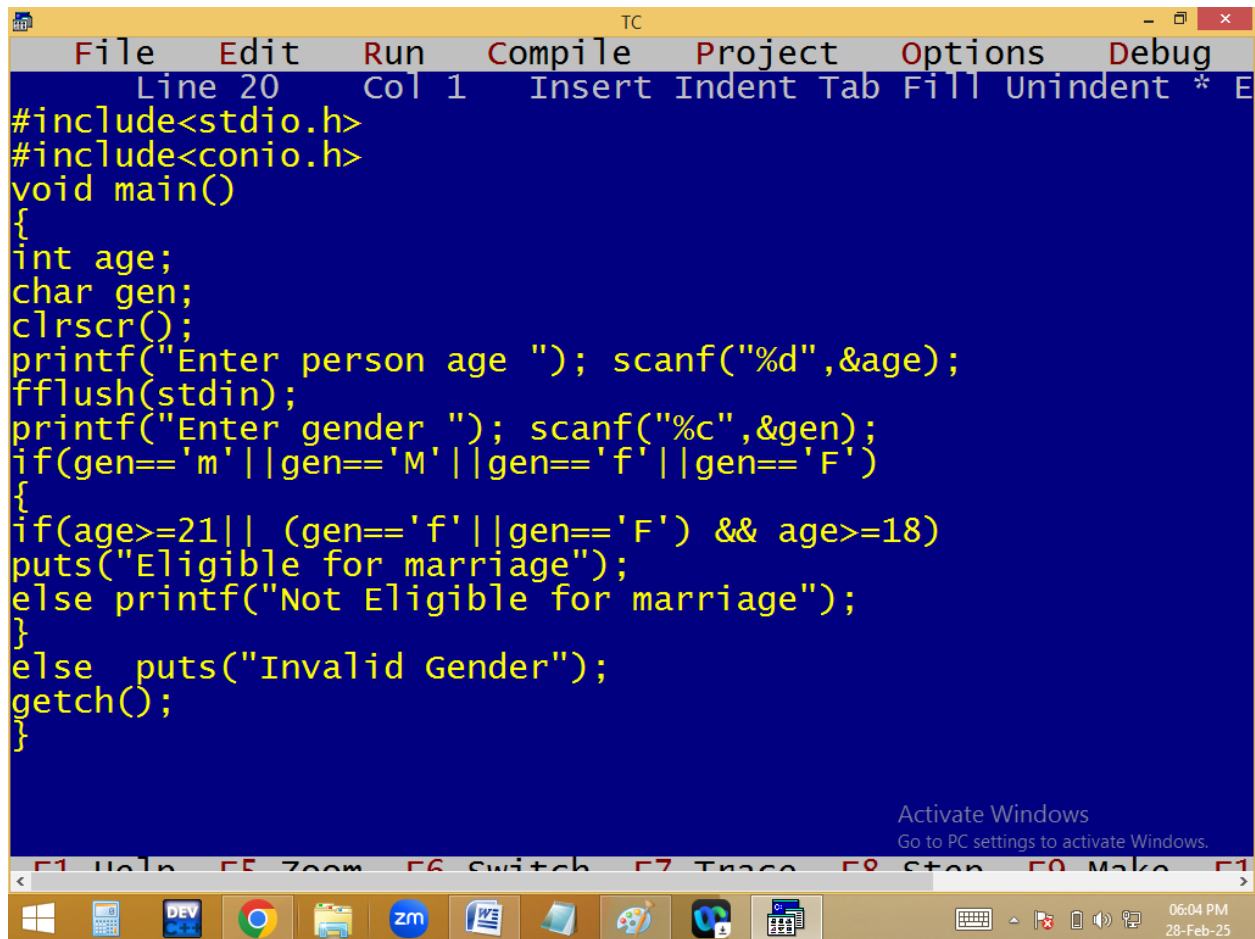
At the bottom of the IDE window, there is a text overlay that says "Activate Windows" and "Go to PC settings to activate Windows." Below the IDE window is the Windows taskbar, which includes icons for the Start menu, Task View, File Explorer, Microsoft Edge, Zoom, Word, and other applications. The system tray on the right shows the time as 05:59 PM and the date as 28-Feb-25.











The image shows a screenshot of the Turbo C++ (TC) IDE. The window title is "TC". The menu bar includes "File", "Edit", "Run", "Compile", "Project", "Options", and "Debug". The status bar at the top indicates "Line 20", "Col 1", and "Insert Indent Tab Fill Unindent \* E". The main editing area has a blue background and contains the following C code:

```
#include<stdio.h>
#include<conio.h>
void main()
{
    int age;
    char gen;
    clrscr();
    printf("Enter person age "); scanf("%d",&age);
    fflush(stdin);
    printf("Enter gender "); scanf("%c",&gen);
    if(gen=='m' || gen=='M' || gen=='f' || gen=='F')
    {
        if(age>=21 || (gen=='f' || gen=='F') && age>=18)
            puts("Eligible for marriage");
        else printf("Not Eligible for marriage");
    }
    else puts("Invalid Gender");
    getch();
}
```

At the bottom right of the IDE, there is a message: "Activate Windows. Go to PC settings to activate Windows." The Windows taskbar is visible at the bottom, showing icons for Windows, Task View, DEV, Google Chrome, File Explorer, Zoom, Word, and other applications. The system clock in the bottom right corner shows "06:04 PM" and "28-Feb-25".

